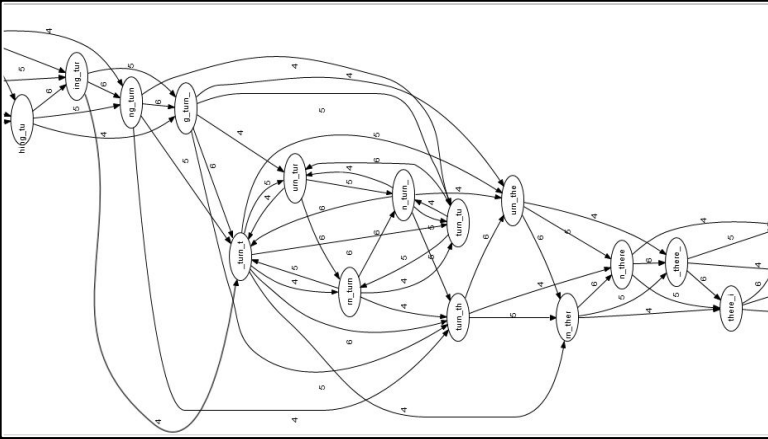
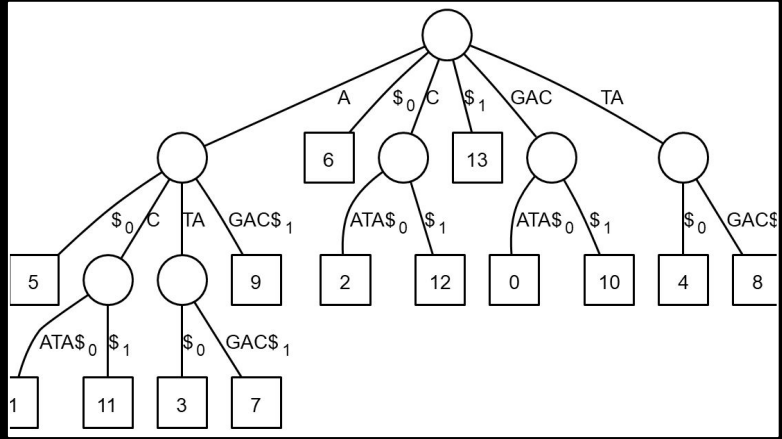
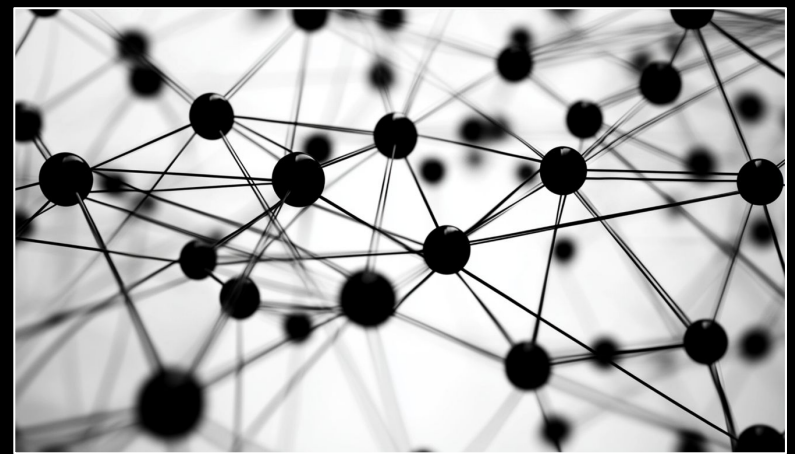
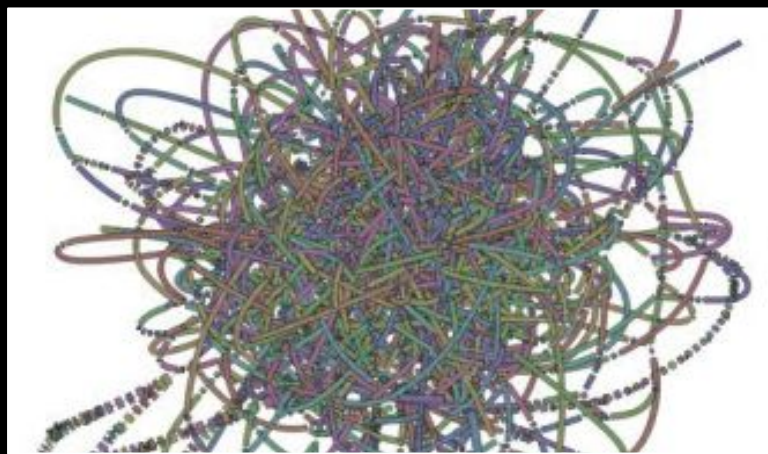
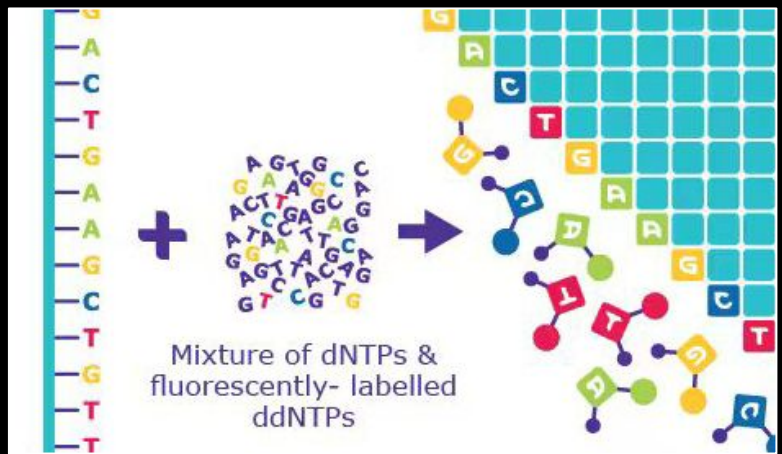




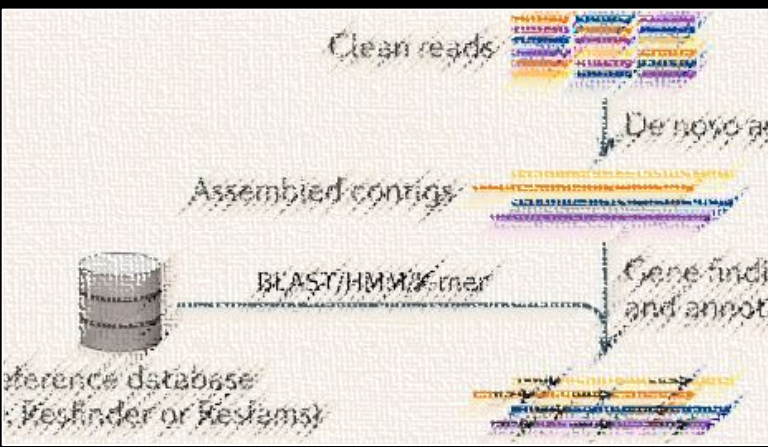
GATTACACAGATTACTGA TTGATGGCGTAA C
 GATTACACAGATTACTGACTTGATGGCGTAAAC
 G TTACACAGATTATTGACTTCATGGCGTAA C
 GATTACACAGATTACTGACTTGATGGCGTAA C
 GATTACACAGATTACTGACTTGATGGCGTAA C

↓ ↓ ↓ ↓ ↓

GATTACACAGATTACTGACTTGATGGCGTAA C



Module 2



Assembly

Module	Week	Date	Topic	Deadlines
<i>Introduction</i>	1	2026-01-07	Overview & Life at Resolution	
	2	2026-01-12	Pathway & Central Dogma	
		2026-01-14	Molecular Evolution	
<i>Homology</i>	3	2026-01-19	Snow Day	
		2026-01-21	Sequence Representation	
	4	2026-01-26	Snow Day	
		2026-01-28	Sequence Alignment: Scoring	
	5	2026-02-02	Sequence Alignment: Dynamic Programming	
		2026-02-04	Homology Search: BLAST	
	6	2026-02-09	Homology Search: Burrows-Wheeler Transform	Assignment 1 Release
		2026-02-11	Hidden Markov Models / Multiple Sequence Alignment	
<i>Winter Study Break</i>	7	2026-02-16	No Lecture	
		2026-02-18	No Lecture	
<i>Assembly</i>	8	2026-02-23	Snow Day	Assignment 1 Due
		2026-02-25	Assembly: Overlap Layout Consensus	
	9	2026-03-02	Assembly: de Bruijn Graphs	Assignment 2 Release
Phylogenetics		2026-03-04	Introduction to Phylogenetics	
	10	2026-03-09	Distance Inference	Assignment 2 Due
		2026-03-11	Maximum Likelihood Inference	Paper Selection Deadline
	11	2026-03-16	Bayesian Inference	Assignment 3 Release
		2026-03-18	Statistical Support & Testing	
	12	2026-03-23	Pangenomics & Outbreaks	Assignment 3 due
		2026-03-25	Phylodynamics	
Machine Learning	13	2026-03-30	Introduction to Machine Learning for Biological Data	
		2026-04-01	Feature Representations	Assignment 4 Release
	14	2026-04-06	Oral Presentations	
		2026-04-08	Oral Presentations	Assignment 4 & Paper Review Due

Module Summary

- The DNA molecules we want to sequence can be **long** (>>> 1 million nucleotides)
- DNA sequencing technology is great for this, but:
 - Most methods only give us short pieces
 - Merging these short pieces is **not trivial**
 - Sequencing can be error prone
- We need sequence **assembly algorithms** to handle this
 - Typically graph based: overlap-consensus (today), *de Bruijn* (next)

Slides adapted from Rob Beiko who adapted the slides from me who adapted the slides from Ben Langmead's Teaching Materials

(www.langmead-lab.org/teaching.html)

Assembly: Overlap Layout Consensus

- or -

Genomic Frustration == Algorithmic Opportunities

Overview

- DNA sequencing technologies
- The assembly problem
- Graph representations
- The overlap-layout-consensus method

What Do You Want to Sequence?

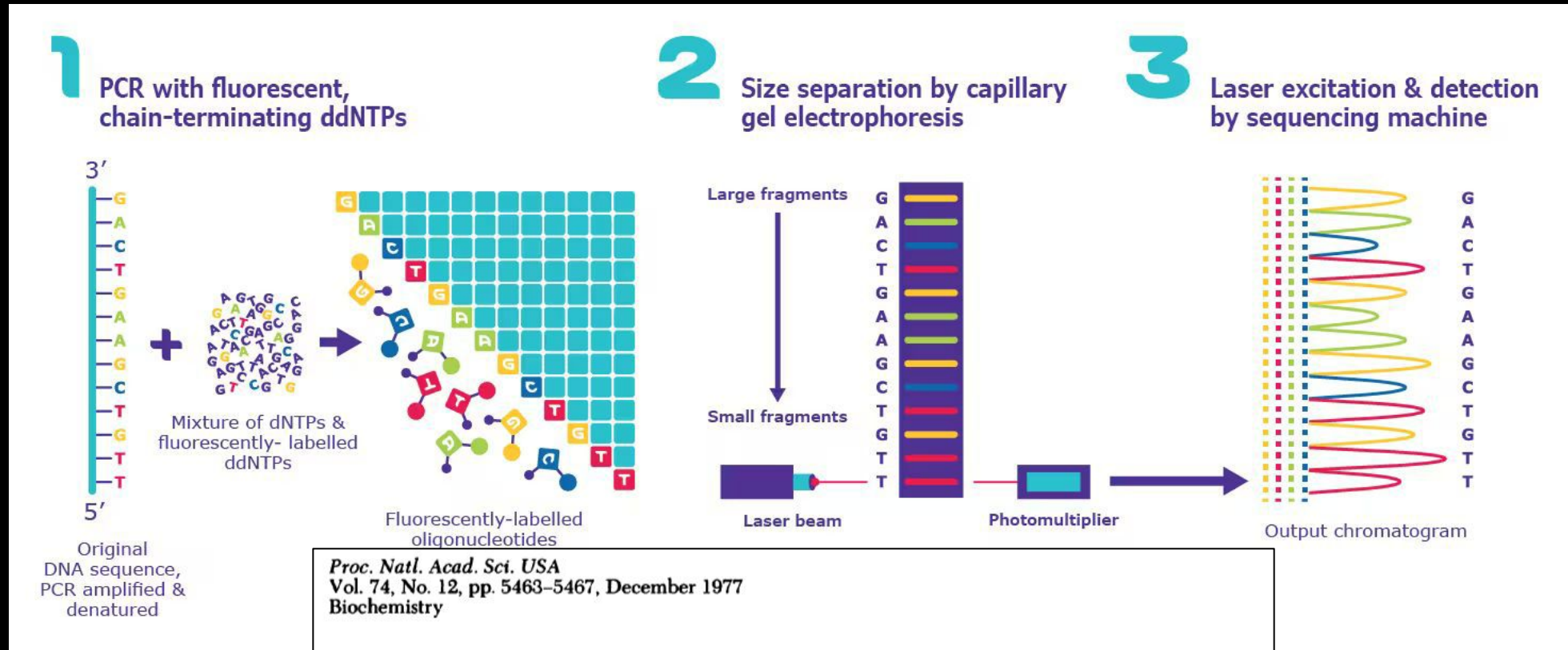
- One gene?
- A few specific genes?
- A section of a chromosome?
- An entire chromosome?
- The entire genome?
- Just the parts that are being transcribed?

- AND -

- How much money do you have?
- How accurate does the sequence need to be?

Great moments in DNA Sequencing:

#1 - Sanger chain-termination sequencing (1977)



Proc. Natl. Acad. Sci. USA
Vol. 74, No. 12, pp. 5463-5467, December 1977
Biochemistry

DNA sequencing with chain-terminating inhibitors

(DNA polymerase/nucleotide sequences/bacteriophage ϕ X174)

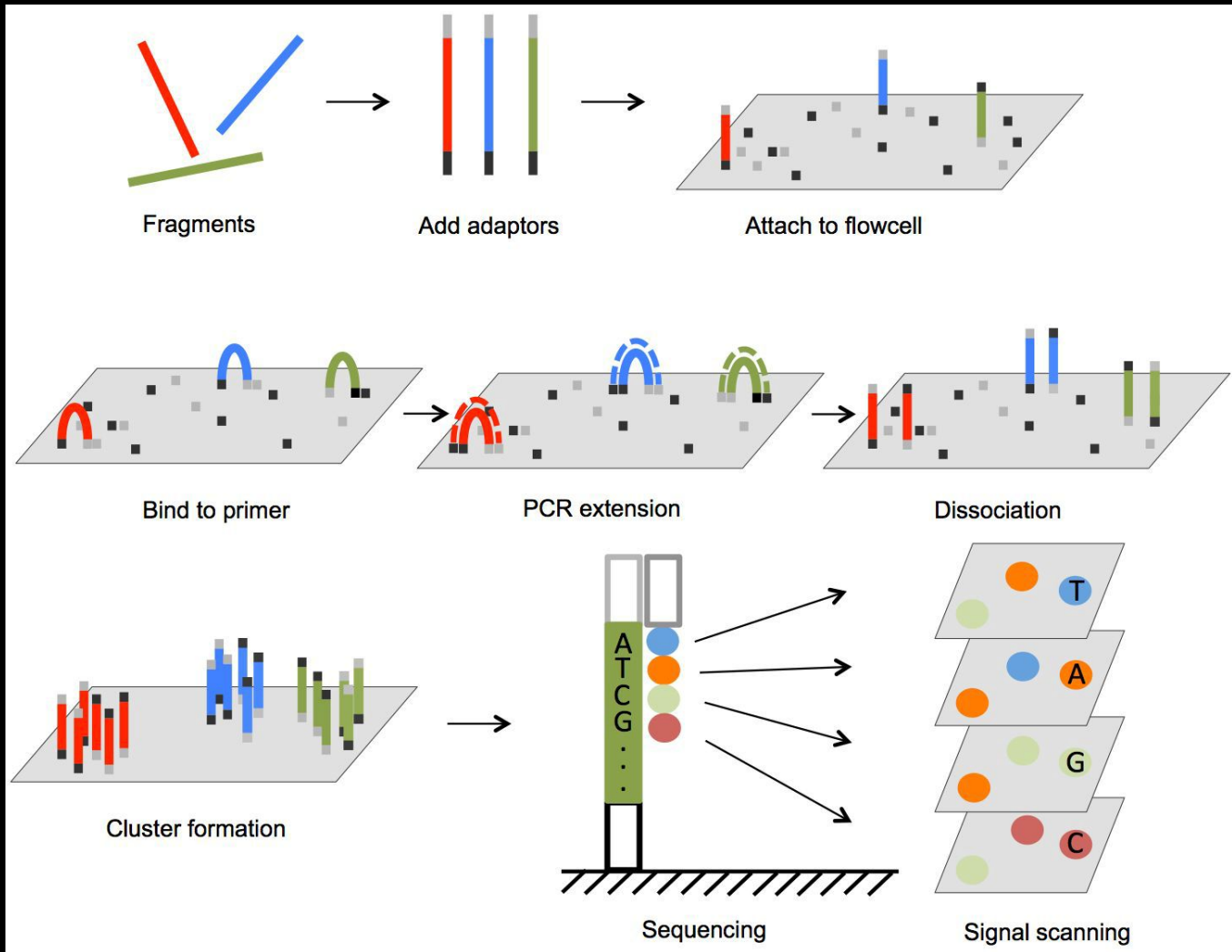
F. SANGER, S. NICKLEN, AND A. R. COULSON

Medical Research Council Laboratory of Molecular Biology, Cambridge CB2 2QH, England

Contributed by F. Sanger, October 3, 1977

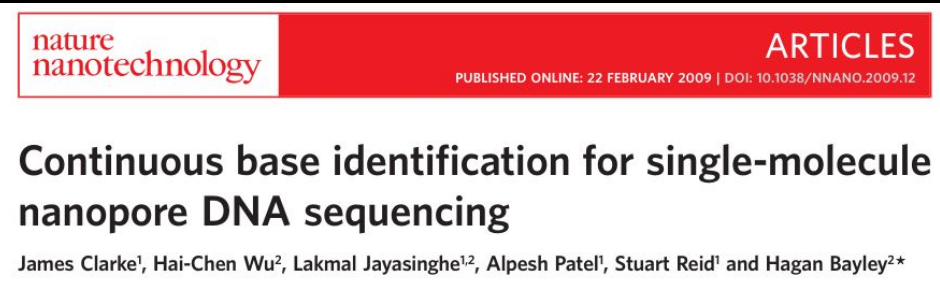
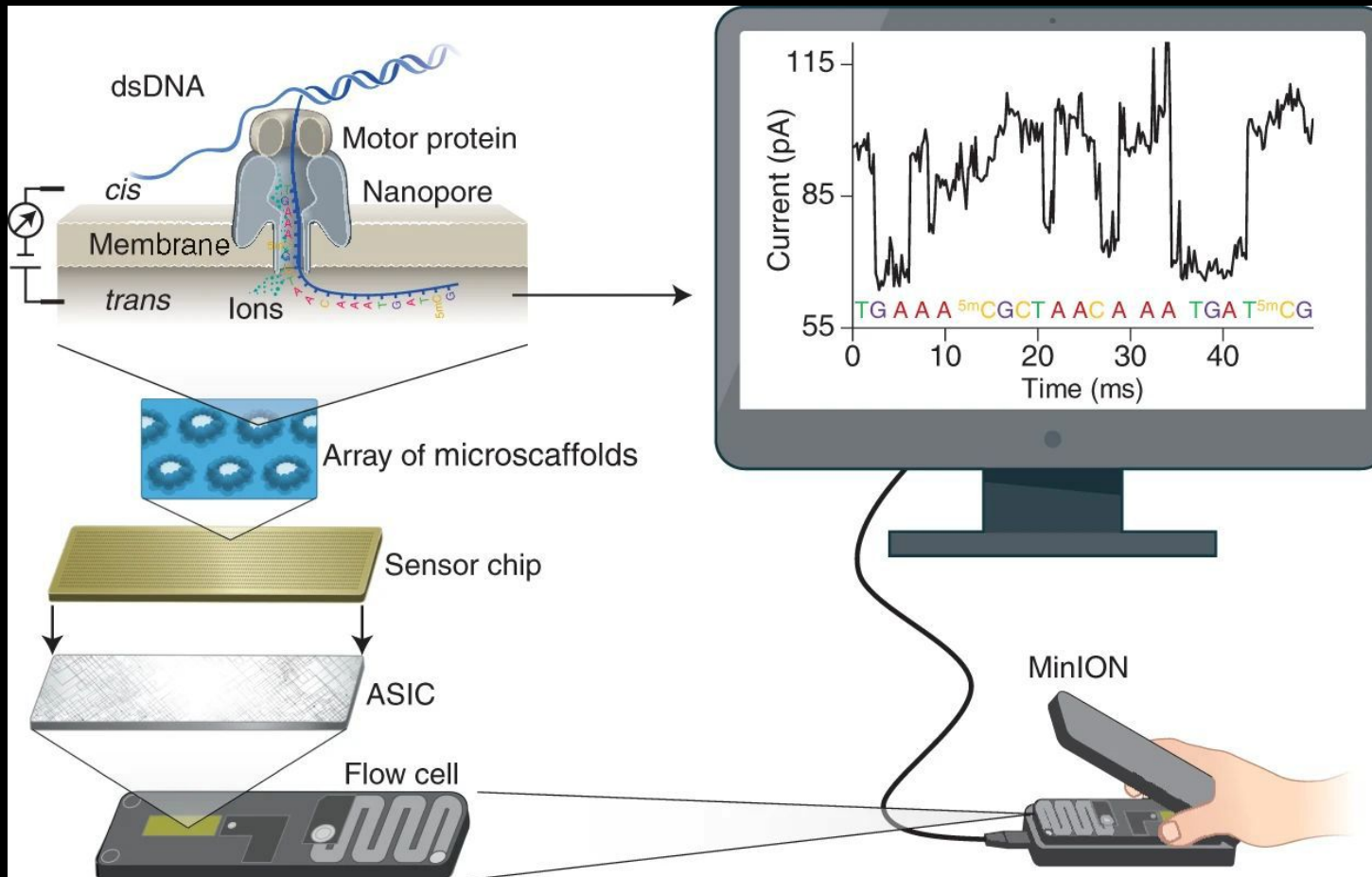
Great moments in DNA Sequencing:

#2 – Sequencing by synthesis (<2008)



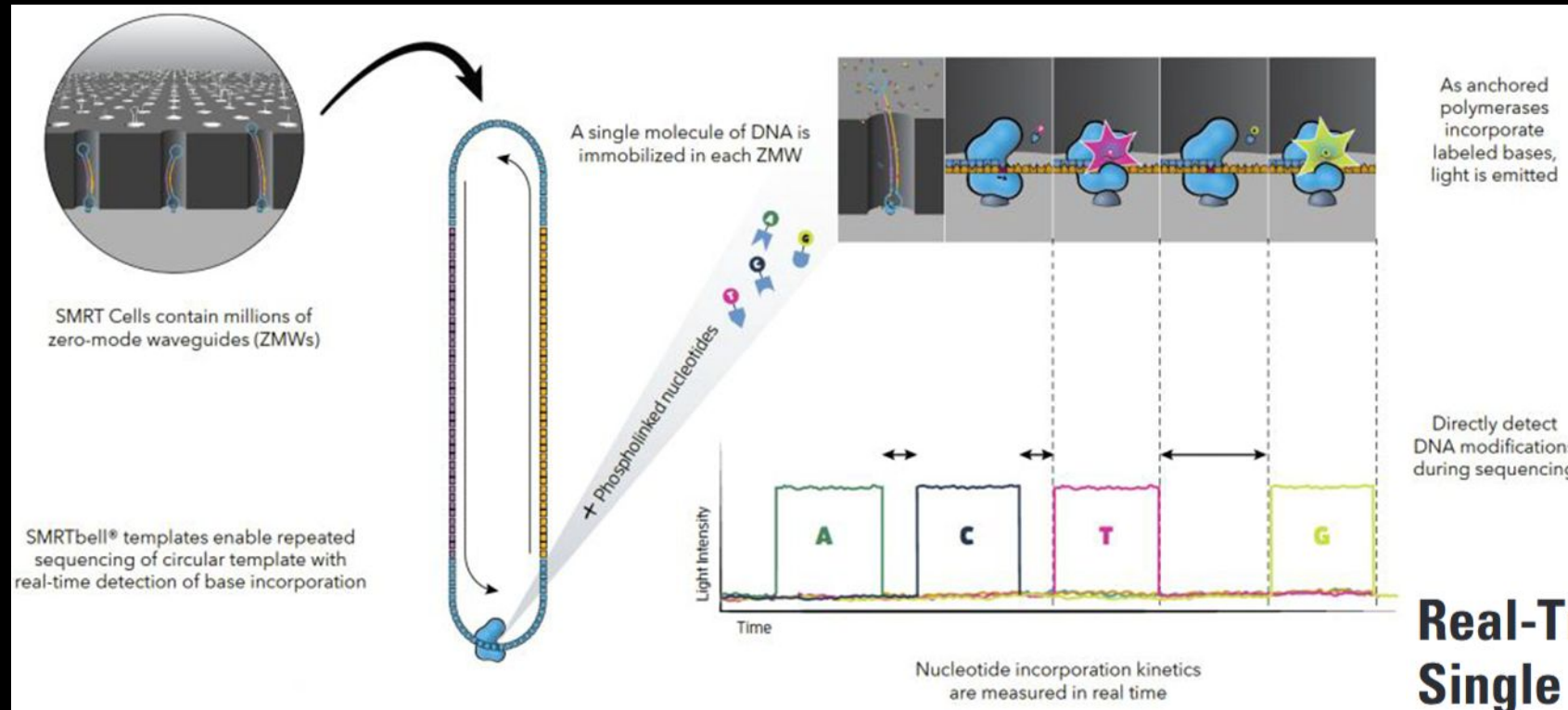
Great moments in DNA Sequencing:

#3 – Nanopore sequencing (2009)



Great moments in DNA Sequencing:

#4 – Single-molecule real time (SMRT) sequencing (2009)



Real-Time DNA Sequencing from Single Polymerase Molecules

John Eid,* Adrian Fehr,* Jeremy Gray,* Khai Luong,* John Lyle,* Geoff Otto,* Paul Peluso,* David Rank,* Primo Baybayan, Brad Bettman, Arkadiusz Bibillo, Keith Bjornson, Bidhan Chaudhuri, Frederick Christians, Ronald Cicero, Sonya Clark, Ravindra Dalal, Alex deWinter, John Dixon, Mathieu Foquet, Alfred Gaertner, Paul Hardenbol, Cheryl Heiner, Kevin Hester, David Holden, Gregory Kearns, Xiangxu Kong, Ronald Kuse, Yves Lacroix, Steven Lin, Paul Lundquist, Congcong Ma, Patrick Marks, Mark Maxham, Devon Murphy, Insil Park, Thang Pham, Michael Phillips, Joy Roy, Robert Sebra, Gene Shen, Jon Sorenson, Austin Tomaney, Kevin Travers, Mark Trulson, John Vieceli, Jeffrey Wegener, Dawn Wu, Alicia Yang, Denis Zaccarin, Peter Zhao, Frank Zhong, Jonas Korlach,† Stephen Turner†

These (+ several others) are still in use

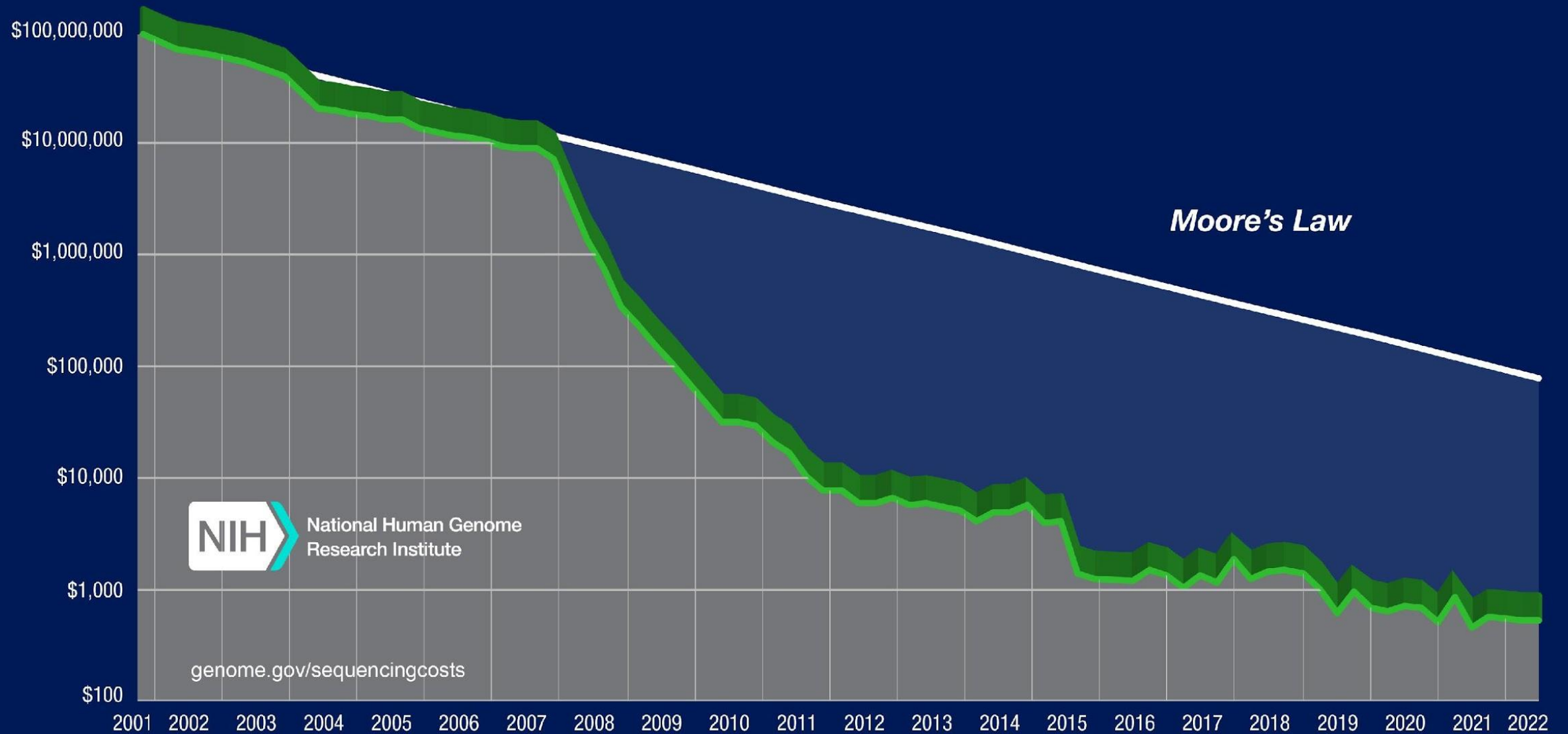
Why?

- Cost per base
- Accuracy
- Length of sequences produced at a time

all differ.

(Plus: legacy reasons)

Cost per Human Genome



The Problem

- DNA sequencing techniques give you sequence **reads**. These are typically shorter (potentially much shorter!) than the DNA molecule you'd like to sequence
- Sanger sequencing: ~1000 bp (low error rate)
- Illumina sequencing: 150-300 bp (lowish error rate)
- Oxford Nanopore: Varies widely (record is over 2 million bp!)
- PacBio SMRT: “15,000 to 20,000 base pairs or more”
(both moderate to high error rates naively!)

Shotgun Sequencing: Fragmentation

Original population of DNA sequences:

GGCGTCTATATCTCGGCTCTAGGCCCTCATTTTTT
GGCGTCTATATCTCGGCTCTAGGCCCTCATTTTTT
GGCGTCTATATCTCGGCTCTAGGCCCTCATTTTTT
GGCGTCTATATCTCGGCTCTAGGCCCTCATTTTTT

Fragmented sequences:

GGC GTCTA TATCTCGG CTCTAGGCCCTC ATTTTTT
G GCGTCTATAT CTCGGCTC TAGGCCCT CATTTTTT
GGCGTCTAT ATC TCGGCTCT AGGCCCTCATT TTTT
GGCGTCT ATA TCTCGG CTCTAGGCCCTCATTTT TT

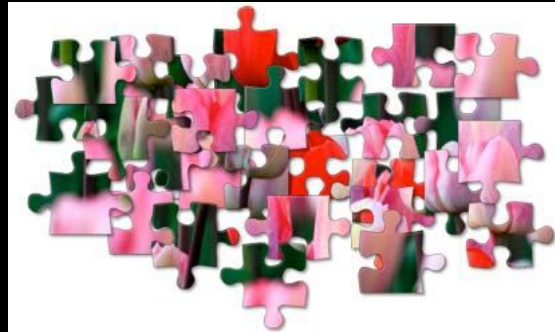
Sequence a subset of fragments **at random**

Assembling Shotgun Sequences

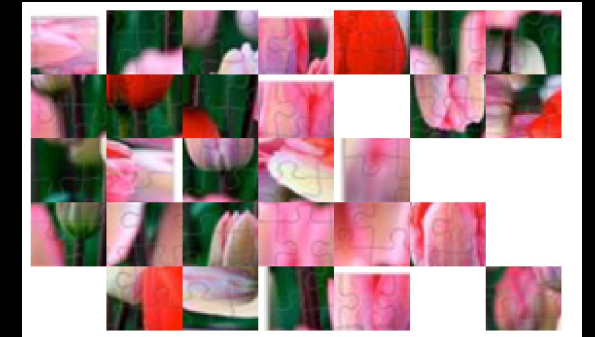
Input DNA



Small pieces



Reassemble into correct full-length sequence

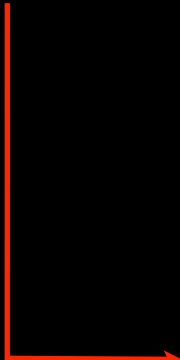


Assembling the Sequence Fragments

Reconstruct this

From these

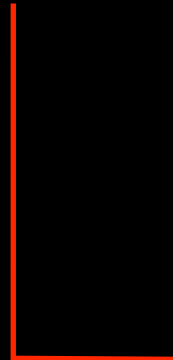
CTAGGCCCTCAATTTTTTT
CTCTAGGCCCTCAATTTTT
GGCTCTAGGCCCTCATTTTTT
CTCGGCTCTAGCCCCTCATTTT
TATCTCGACTCTAGGCCCTCA
TATCTCGACTCTAGGCC
TCTATATCTCGGCTCTAGG
GGCGTCTATATCTCG
GGCGTCGATATCT
GGCGTCTATATCT
GGCGTCTATATCTCGGCTCTAGGCCCTCATTTTTT



Easy!

Assembling the Sequence Fragments

Reconstruct this



????????????????????????????????????

CTAGGCCCTCAATTTTTTT
CTCTAGGCCCTCAATTTTT
GGCTCTAGGCCCTCATTTTT
CTCGGCTCTAGCCCCTCATTTT
TATCTCGACTCTAGGCCCTCA
TATCTCGACTCTAGGCC
TCTATATCTCGGCTCTAGG
GGCGTCTATATCTCG
GGCGTCGATATCT
GGCGTCTATATCT

From
these

maybe not so easy

Coverage

- What is the average number of times each nucleotide in the original DNA is part of a sequenced fragment?

CTAGGCCCTCAATTTT
CTCTAGGCCCTCAATTTT
GGCTCTAGGCCCTCATTTTT
CTCGGCTCTAGCCCCTCATT
TATCTCGACTCTAGGCCCTCA
TATCTCGACTCTAGGCC
TCTATATCTCGGCTCTAGG
GCGTCTATATCTCG
GCGTCGATATCT
GCGTCTATATCT
GCGTCTATATCTCGGCTCTAGGCCCTCATTTTT

Coverage = 5

Coverage

- What is the average number of times each nucleotide in the original DNA is part of a sequenced fragment?

```
CTAGGCCCTCAATTTTT
CTCTAGGCCCTCAATTTTT
GGCTCTAGGCCCTCATTTTTT
CTCGGCTCTAGCCCCTCATT
TATCTCGACTCTAGGCCCTCA
TATCTCGACTCTAGGCC
TCTATATCTCGGCTCTAGG
GGCGTCTATATCTCG
GGCGTCGATATCT
GGCGTCTATATCT
GGCGTCTATATCTCGGCTCTAGGCCCTCATTTTTT
```

Coverage = 3

Coverage

- Average coverage = # of nucleotides sequenced x total length

Total sequenced = 177

Total length = 36

Average coverage = 4.92

CTAGGCCCTCAATTTTT
CTCTAGGCCCTCAATTTTT
GGCTCTAGGCCCTCATTTTTT
CTCGGCTCTAGCCCCTCATT
TATCTCGACTCTAGGCCCTCA
TATCTCGACTCTAGGCC
TCTATATCTCGGCTCTAGG
GGCGTCTATATCTCG
GGCGTCGATATCT
GGCGTCTATATCT
GGCGTCTATATCTCGGCTCTAGGCCCTCATTTTTT

Coverage

- Coverage is not uniform, so a given coverage level C will have some genomic regions with a depth $> C$, and others with depth $< C$ (including some that may not be covered at all!)
- **IF coverage is unbiased**, then read depth will fit a Poisson distribution

Coverage

Formula for % of genome sequenced at least once:

$$P = 1 - e^{-C}$$

Percentage $\nearrow$ P $\nwarrow$ Euler's constant: 2.718... $\nwarrow$ e^{-C} $\nwarrow$ Average coverage C

$$C = 1: P \approx 63.2\%$$

$$C = 5: P \approx 99.3\%$$

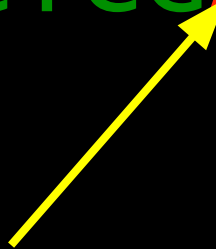
$$C = 10: P \approx 99.995\%$$

Typically, coverage needs to be higher so we can deal with sequencing errors, non-random read depth, assembly weirdness, ...

Overlapping Reads

TCTATATCTCGGCTCTAGG
|||||||
TATCTCGACTCTAGGCC

Mismatches



Overlapping Reads

TCTATATCTCGGCTCTAGG
| | | | | | | | | | | | | |
TATCTCGACTCTAGGCC

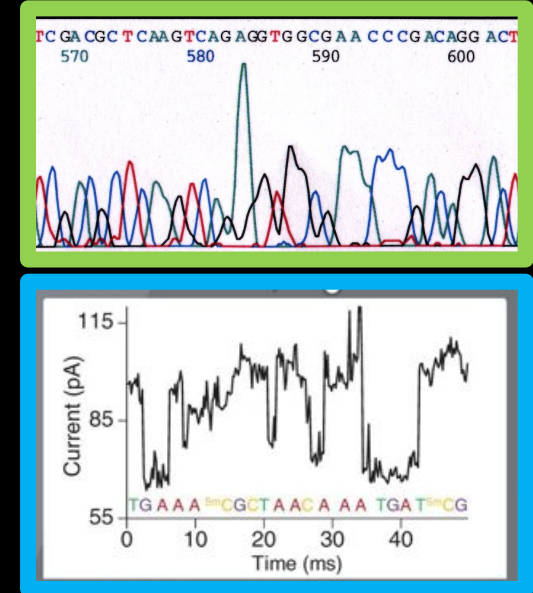
Why mismatches?

- Sequencing errors
- Variation in copies of the source DNA (ploidy, co-occurring strains, repeats)

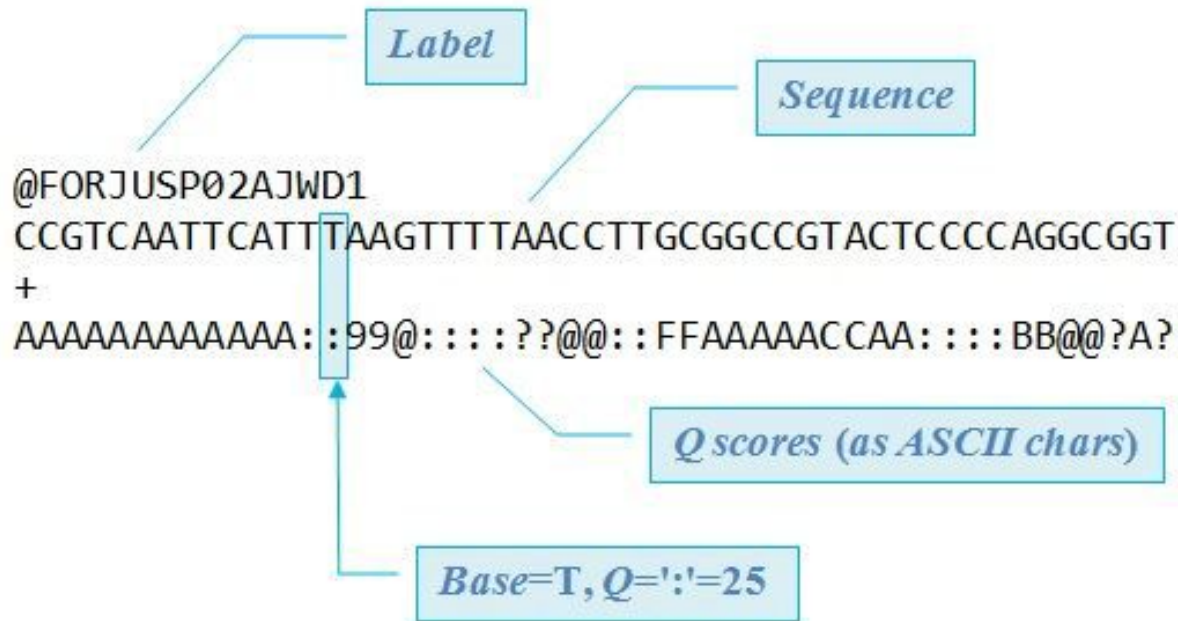


Why Sequencing Errors?

- Misincorporation of a base
- Misreading of a base through **fluorescence**, **pore signal**, etc
- Difficulty with homopolymer runs:
GGGGGG → GGGGGGGG
- Different sequencing technologies have different error rates and error profiles



Capturing measurement error: The FASTQ Format



Quality value Q is an integer representation of the probability p that a corresponding base call is incorrect

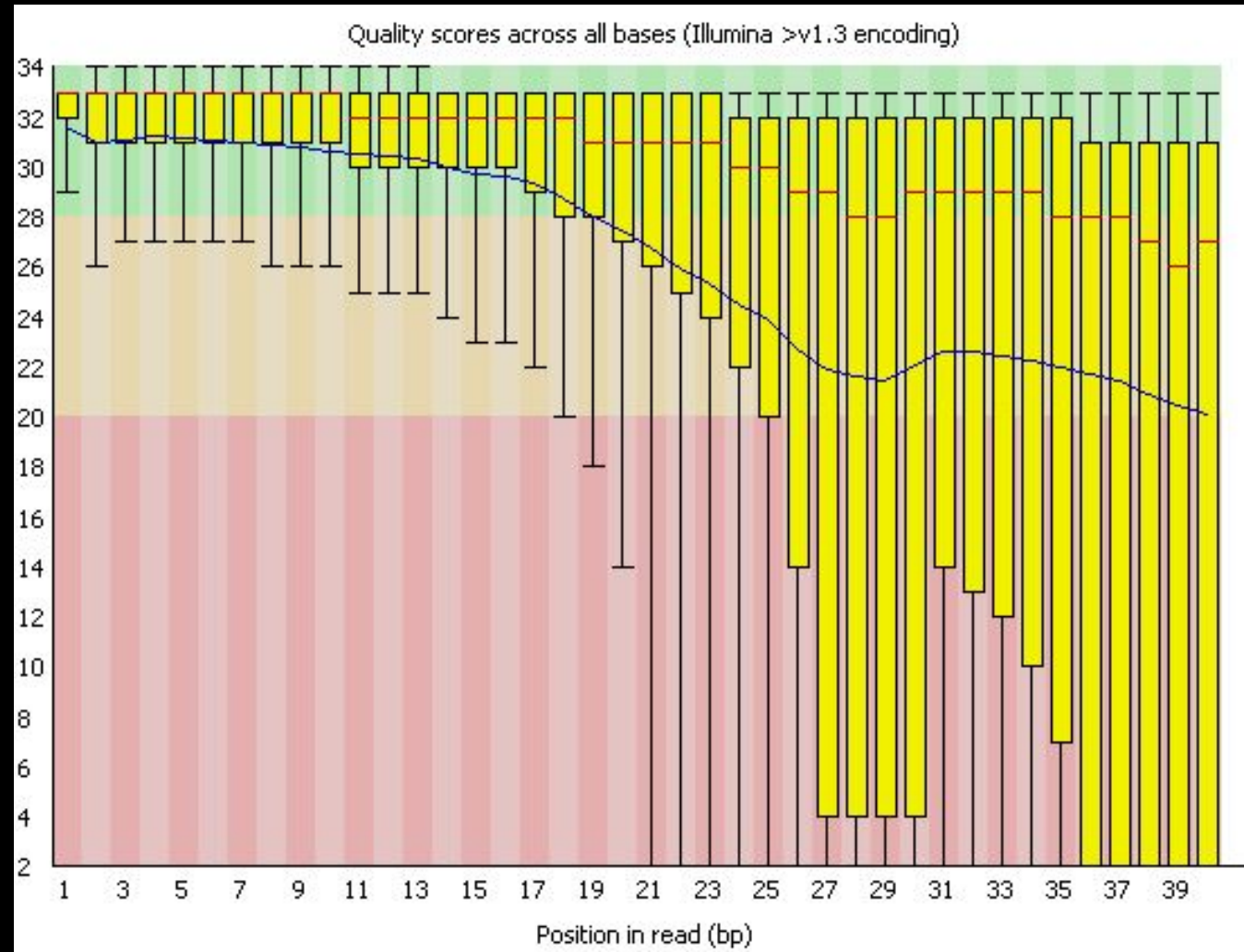
$$Q = -10 \log_{10} P \quad \longrightarrow \quad P = 10^{\frac{-Q}{10}}$$

Phred Quality Score	Probability of incorrect base call	Base call accuracy
10	1 in 10	90%
20	1 in 100	99%
30	1 in 1000	99.9%
40	1 in 10000	99.99%
50	1 in 100000	99.999%

https://www.drive5.com/usearch/manual/fastq_files.html

<https://learn.gencore.bio.nyu.edu/ngs-file-formats/quality-scores/>

Accuracy across a sequence read



First Law of Assembly

If a **suffix** of read A is similar to a **prefix** of read B...

A: TCTATATCTCGGCTCTAGG

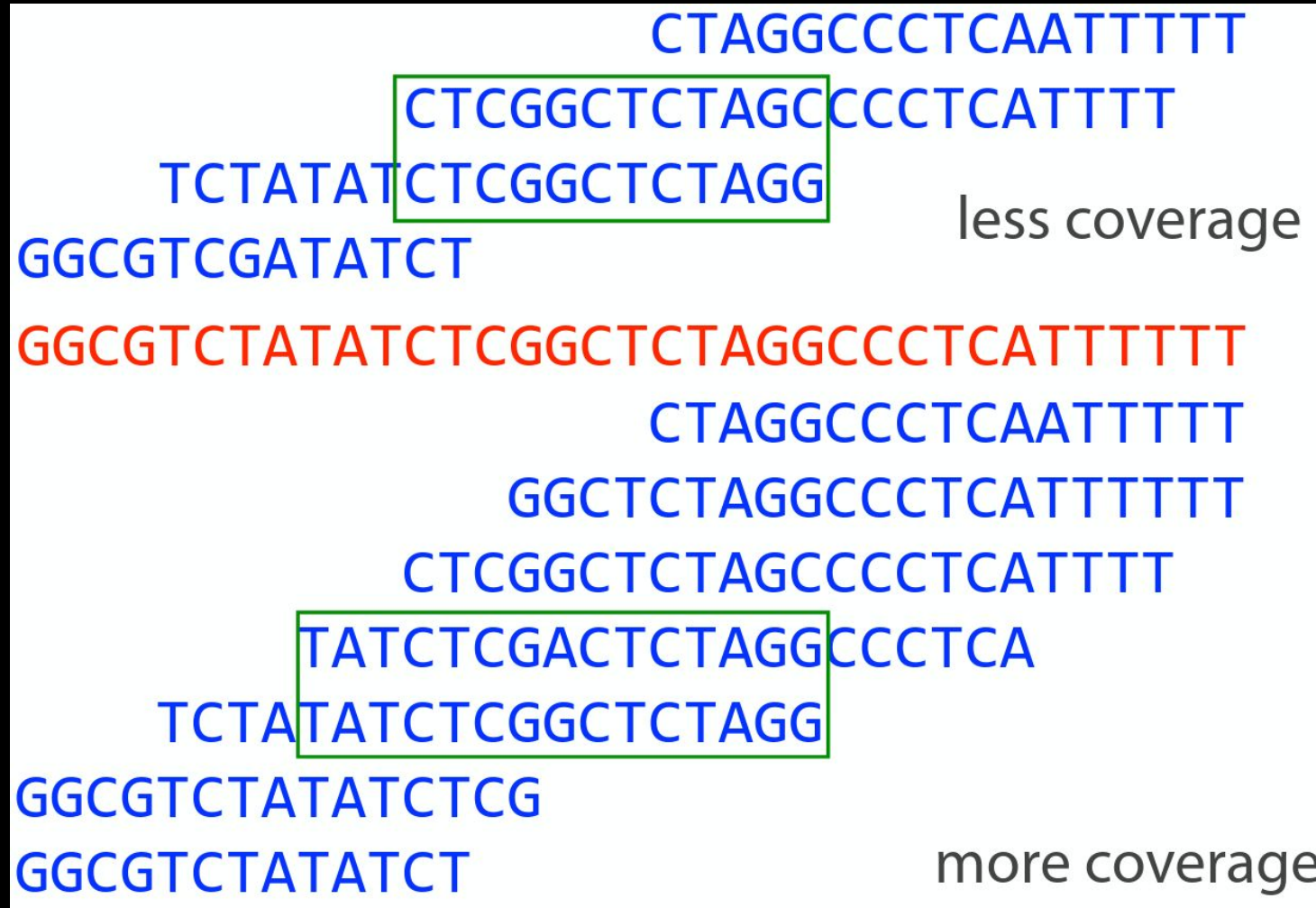
B: TATCTCGACTCTAGGCC

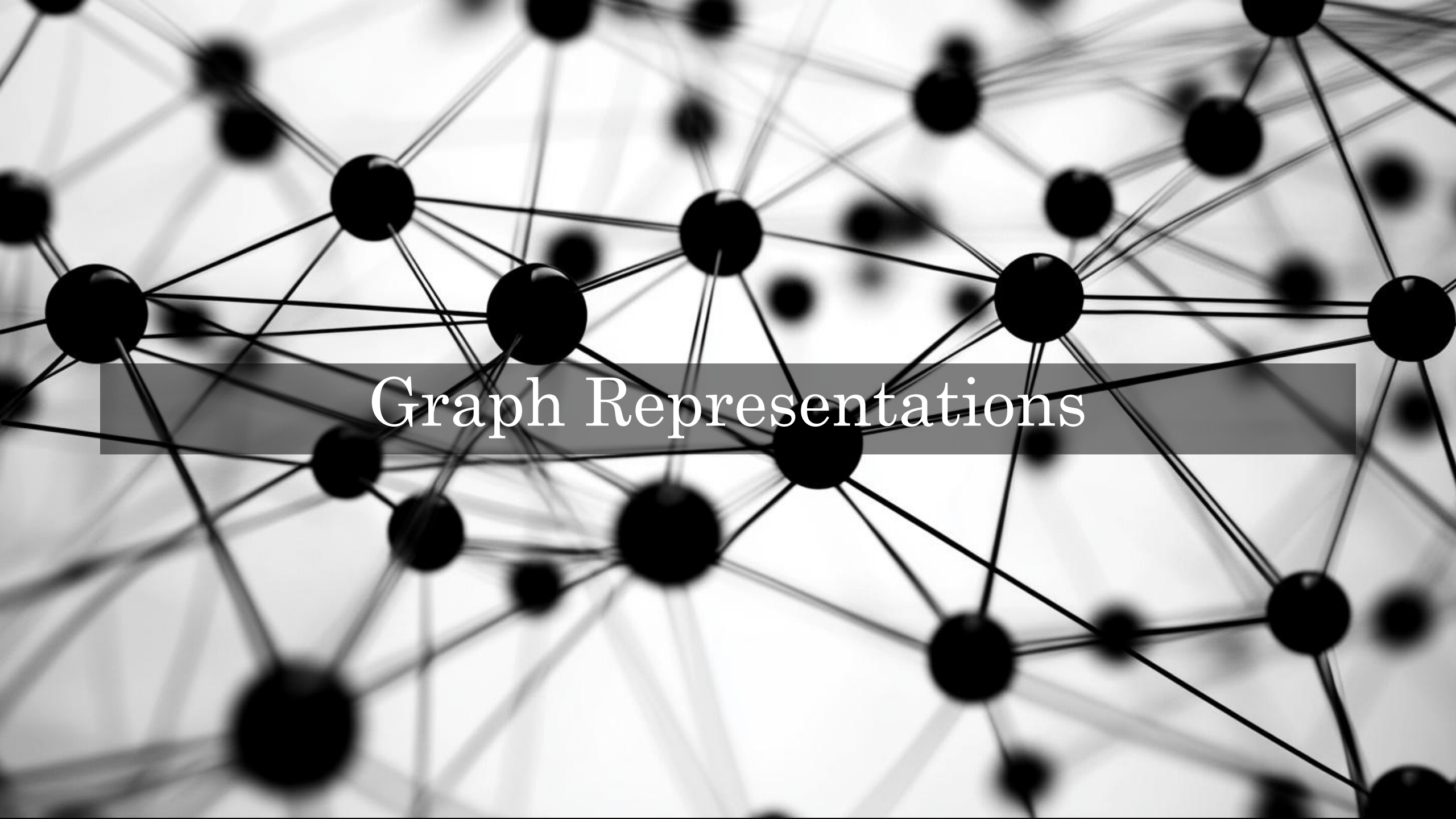
...then A and B might **overlap** in the genome

```
TCTATATCTCGGCTCTAGG
GGCGTCTATATCTCGGCTCTAGGCCCTCATTTTTT
TATCTCGACTCTAGGCC
```

Second Law of Assembly

More coverage leads to more and longer overlaps



The background of the slide is a dense, complex network graph. It consists of numerous black circular nodes of varying sizes, interconnected by a web of thin, dark gray lines representing edges. The nodes are distributed across the frame, with some appearing more prominent than others. The overall effect is a sense of interconnectedness and complexity, typical of network theory or data science visualizations.

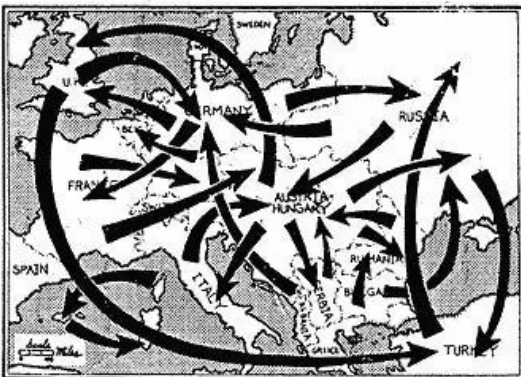
Graph Representations

WAR DECLARED BY ALL

AUSTRIA DECLARES WAR ON SERBIA DECLARES WAR ON
GERMANY DECLARES WAR ON FRANCE DECLARES WAR ON
TURKEY DECLARES WAR ON RUSSIA DECLARES WAR ON
BULGARIA DECLARES WAR ON BRITAIN

OTTOMAN EMPIRE ALMOST DECLARES WAR ON ITSELF
NATIONS STRUGGLE TO REMEMBER ALLIES

From the London and Washington Bureaus, Aug. 4.—After weeks of unbearable tension following the assassination of the heir to the Austro-Hungarian throne, the great European powers have declared war on one another in a bitter struggle for something that is sure to be determined in a confusing string of rivalries and alliances, diplomats, politicians, and military leaders



ASSASSINATION OF ARCHDUKE SPREADS FEAR AT ARCHDUKE CONVENTION

The Hague, Netherlands, Aug. 4.—European archdukes attending their annual convention expressed alarm at the assassination of fellow aristocrat Archduke Franz Ferdinand, whose end came at the hand of a Serbian nationalist in Sarajevo several weeks ago.

Archdukes attending the convention expressed concern for the issue of nobleman safety, once considered a birthright. Most also claimed they had no knowledge of the Serbian demand for independence from Austria. "Could it be that the oppressed minority peoples in our own

AREA DRUNKARD DECLARES WAR ON IRELAND

ALE-HOUSE PEERS FALL IN AS ALLIES

Davenport, Iowa, Aug. 4.—As bloody conflict rages throughout Europe, Orvald Brunvald, a drunkard of many years' experience, declared war on Ireland Friday evening.

A RESOUNDING DECLARATION: "Consarn ever" one of 'em turtles of authoritarian rule, as well!" Brunvald stated over a mug of porter at Quigley's Ale House in Davenport. "I'll lick 'em all to death!" he said. "Filthy potato-eating Papists, they are."



President Wilson, who has vowed to maintain U.S. apathy.

mightily Triple Entente alliance, which no one can really put his finger on, either.

CABLE NOTIFIES WILSON: President Woodrow Wilson received the fateful news via a trans-Atlantic cable com-

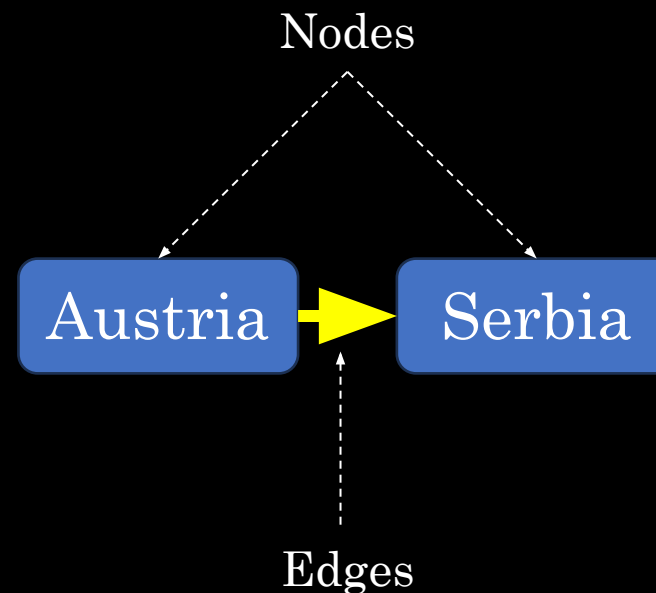
not exist before the June 18th assassination of Austrian Archduke Franz Ferdinand in Sarajevo, but some of the fast-mobilizing nations seem skeptical about crediting his death as the cause of this new and

all European nations who wish to obliterate each other from the face of the Earth."

"WAR TO END ALL EUROPE": American observers are calling this great, worldwide war the "War to End All Europe," as new techno-

The convention, held yearly

Directed Graphs

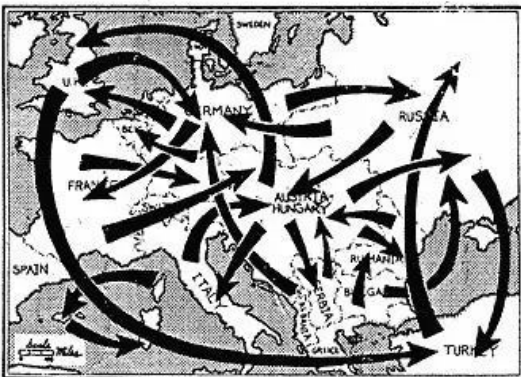


WAR DECLARED BY ALL

AUSTRIA DECLARES WAR ON SERBIA DECLARES WAR ON
 GERMANY DECLARES WAR ON FRANCE DECLARES WAR ON
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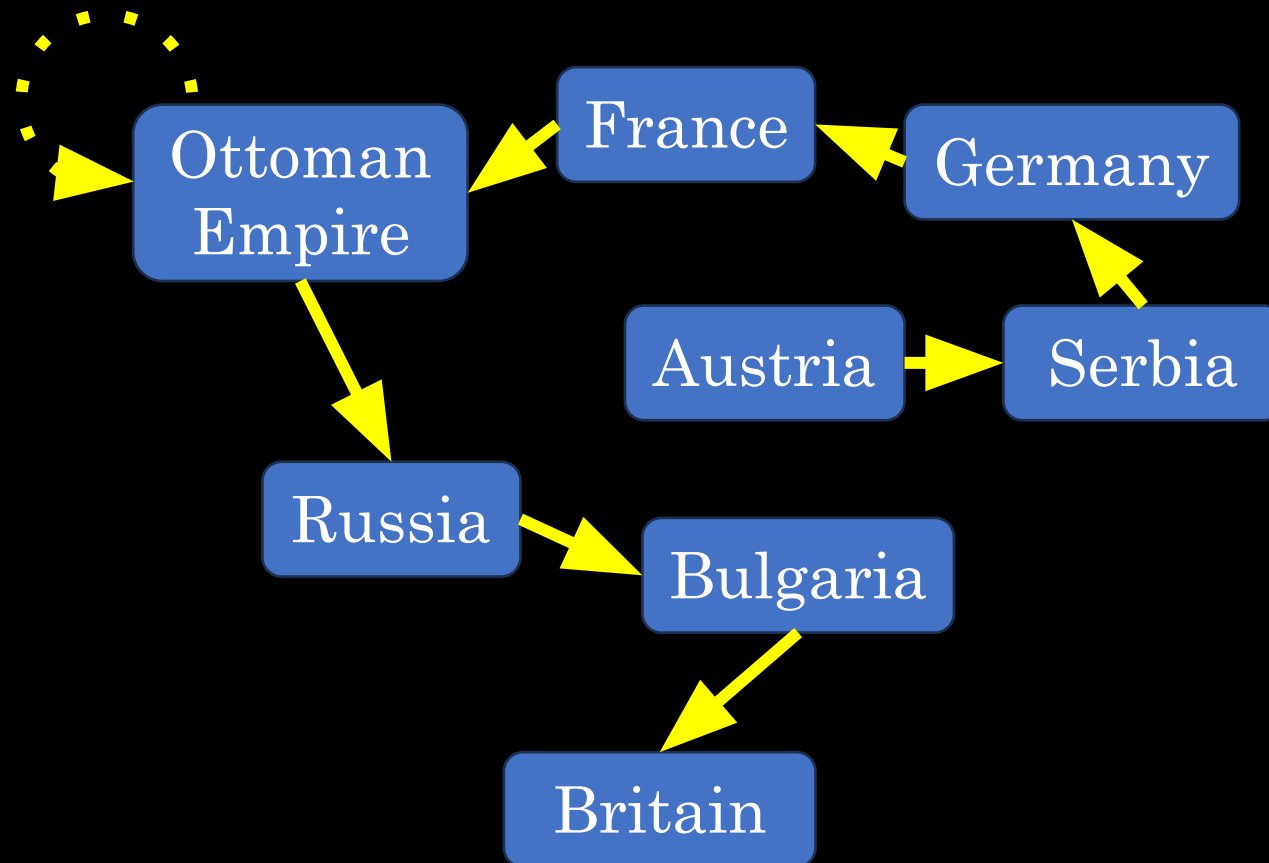
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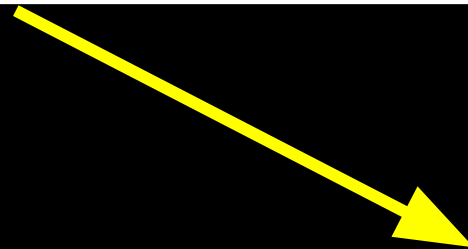
Overlap Graph

Each node is a read

CTCGGGCTCTAGCCCCTCATTTT

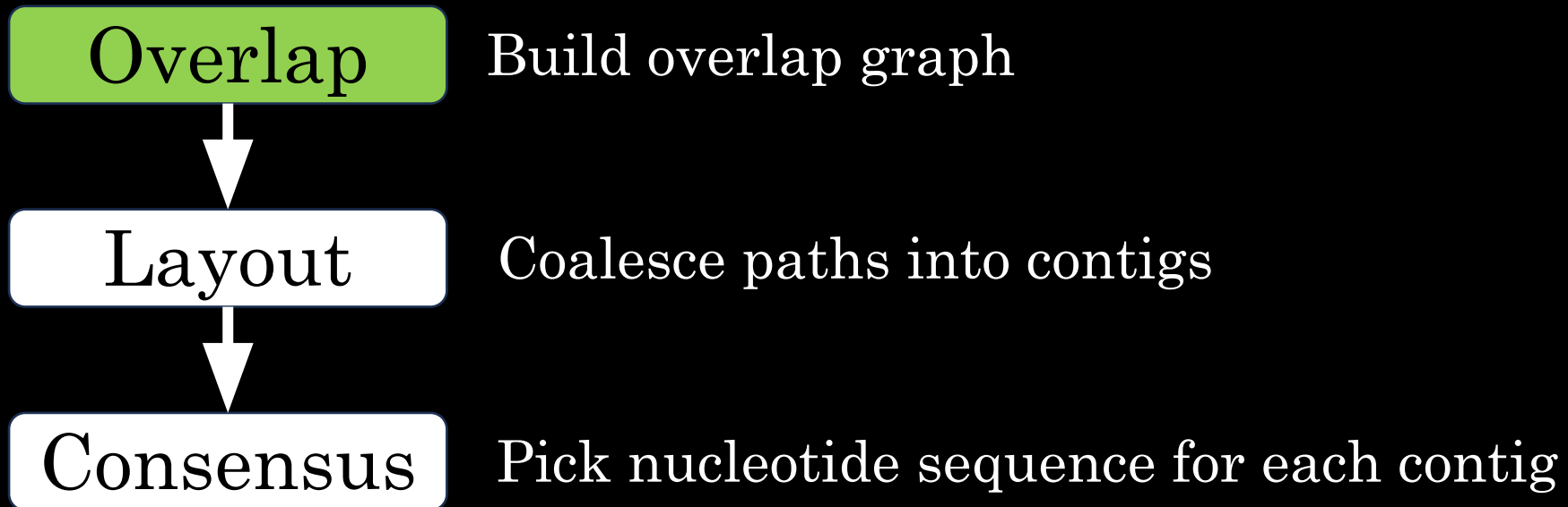
Draw edge $A \rightarrow B$ when suffix of A overlaps prefix of B

CTCGGGCTCTAGCCCCTCATTTT



GGCTCTAGGCCCTCATTTT

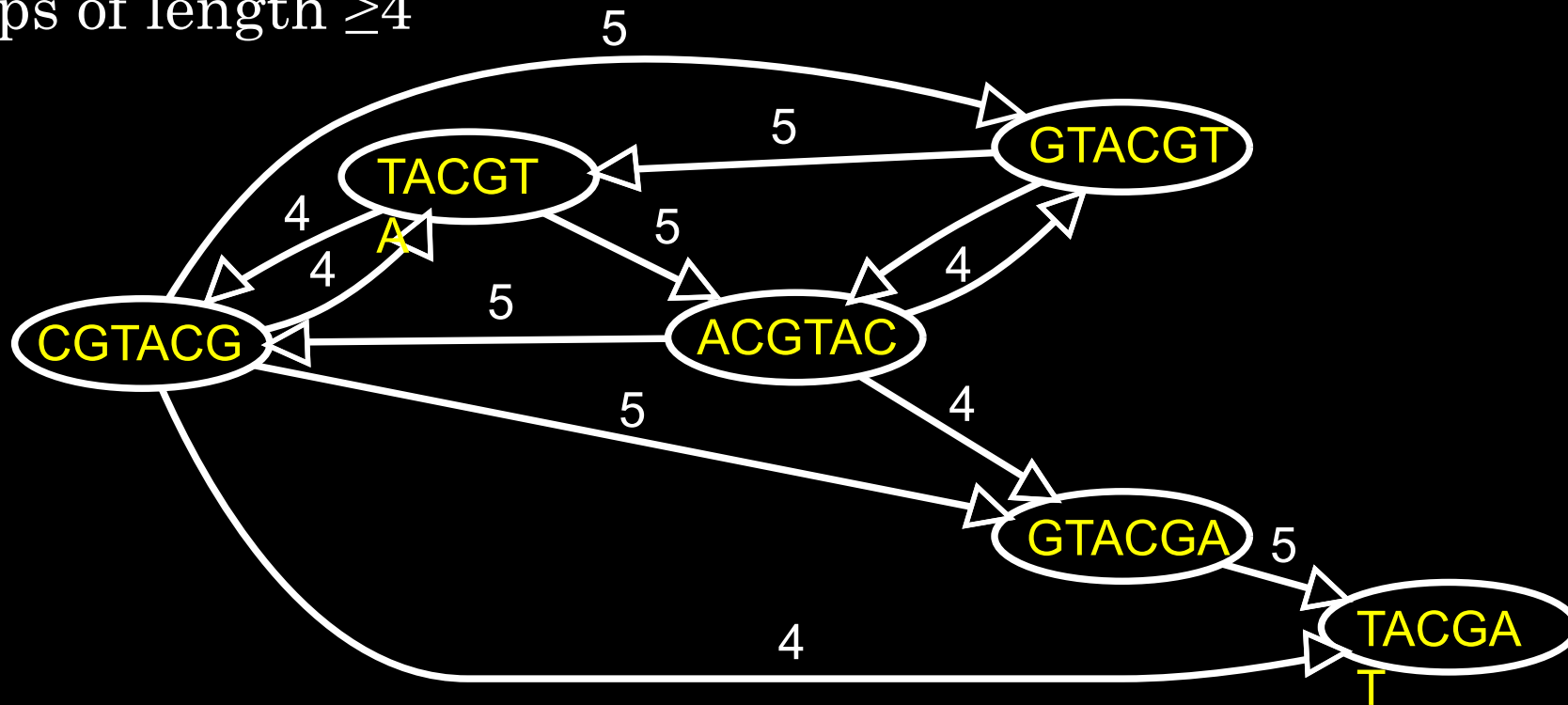
OLC: Building Longer Contiguous Sequences (**contigs**) From Reads



Overlap Graph

Nodes: all 6-mers from GTACGTACGAT

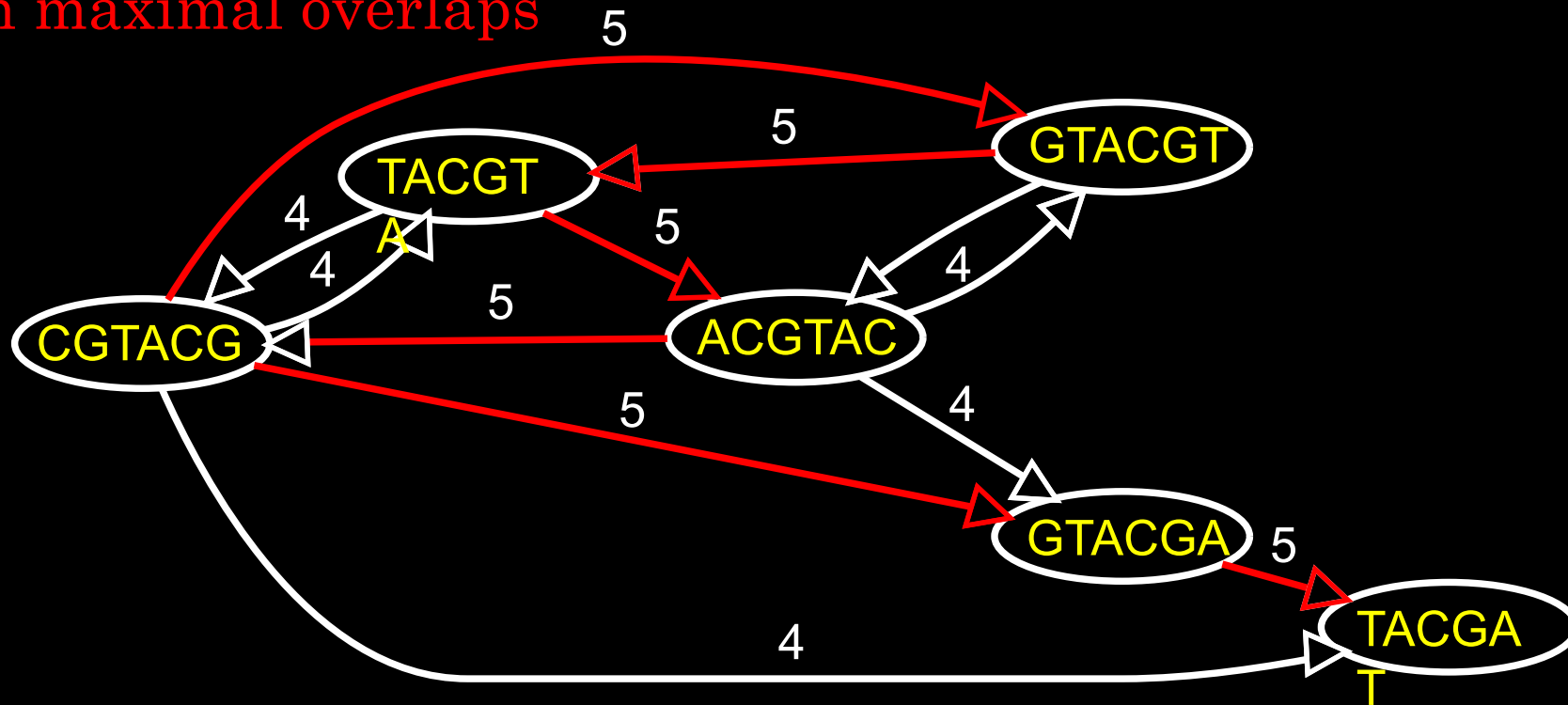
Edges: overlaps of length ≥ 4



Overlap Graph

Nodes: all 6-mers from GTACGTACGAT

Path based on maximal overlaps



Finding Overlaps

Overlap: Suffix of X of length $\geq l$ matches prefix of Y ; l is given

Naïve: Look in X for occurrences of Y 's length- l prefix. Extend matches to the right to confirm whether entire suffix of X matches.

Say $l = 3$

X : CTCTAGGCC
 Y : TAGGCCCTC



Look for this in X

Found it



X : CTCTAGGCC
 Y : TAGGCCCTC

Extend to right; confirm a length-6
prefix of Y matches a suffix of X

X : CTCTAGGCC
 Y : TAGGCCCTC



Finding Overlaps – Suffix Tree

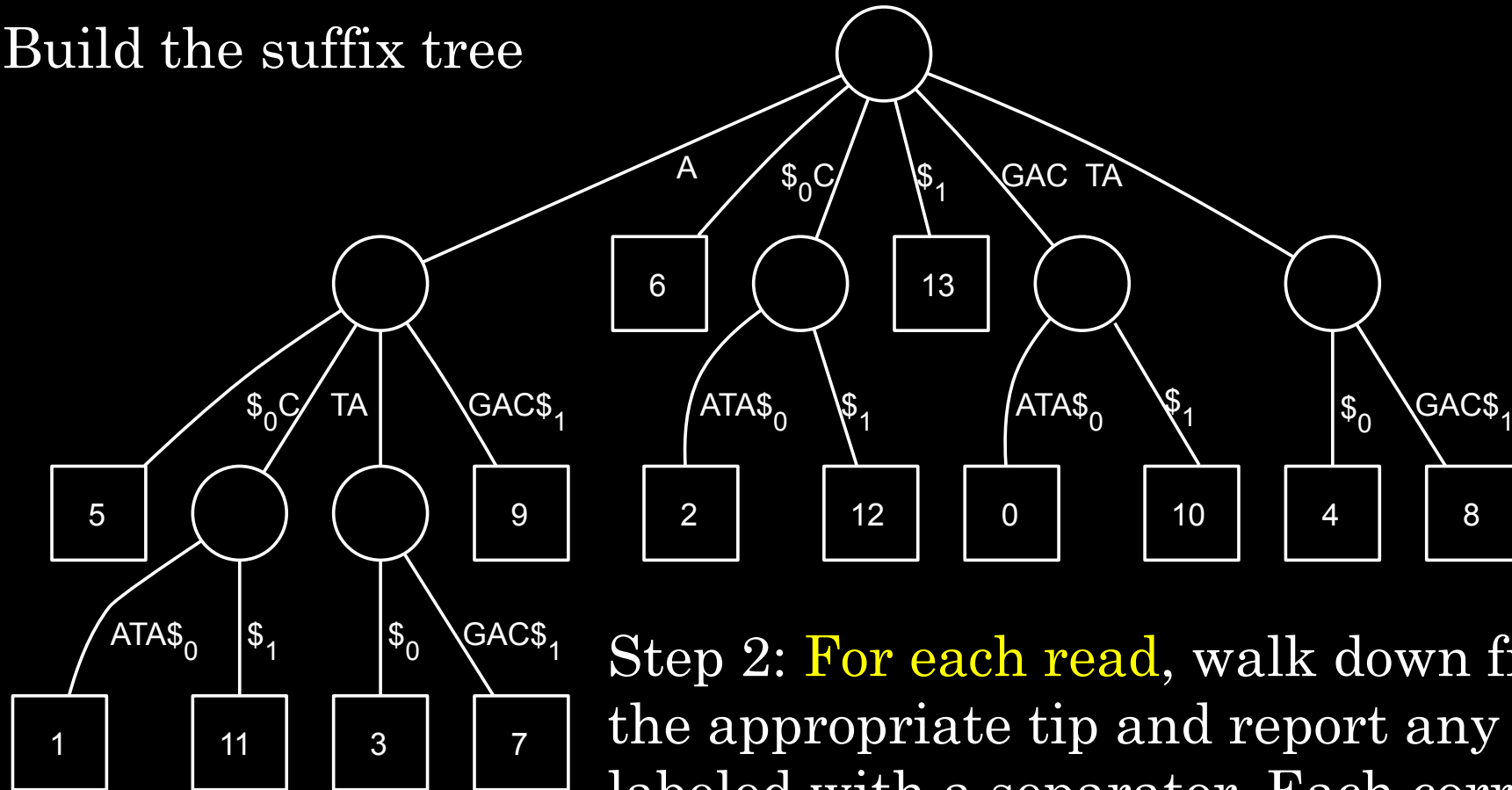
- Given a collection of strings S , for each string X in S ,
- Find all **overlaps** involving a prefix of X and a suffix of another string Y
- Build a suffix tree from all reads; put a terminator character (remember \$?) at the end of each read

Concatenated representation of { “GACATA”, “ATAGAC” }:

GACATA\$₀ATAGAC\$₁

GACATA\$₀ATAGAC\$₁

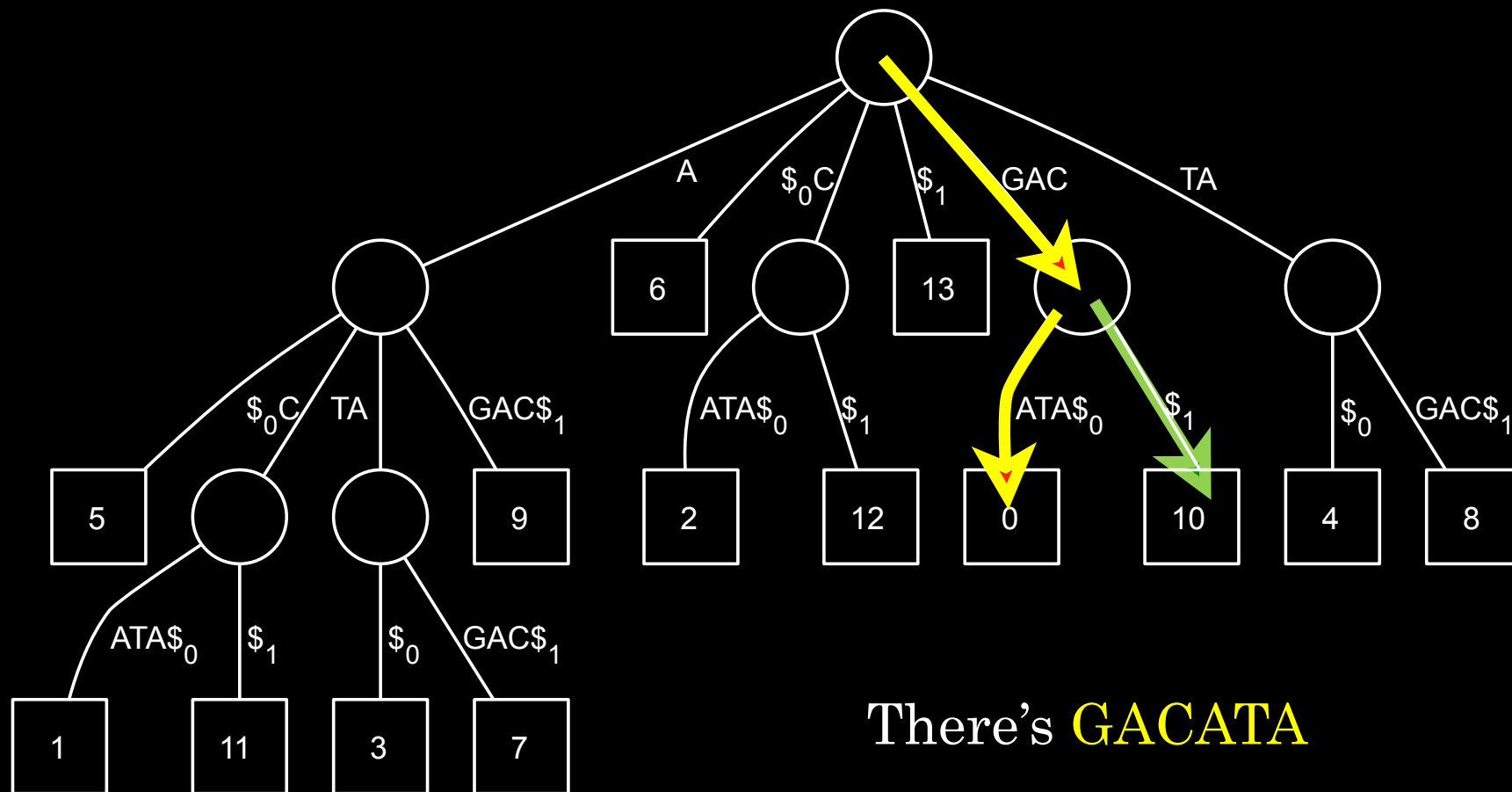
Step 1: Build the suffix tree



Step 2: **For each read**, walk down from the root to the appropriate tip and report any outgoing edge labeled with a separator. Each corresponds to a prefix/suffix match of query read and suffix of another read that ends in the separator.

GACATA\$₀ATAGAC\$₁

Let's start by searching for **GACATA**



There's **GACATA**

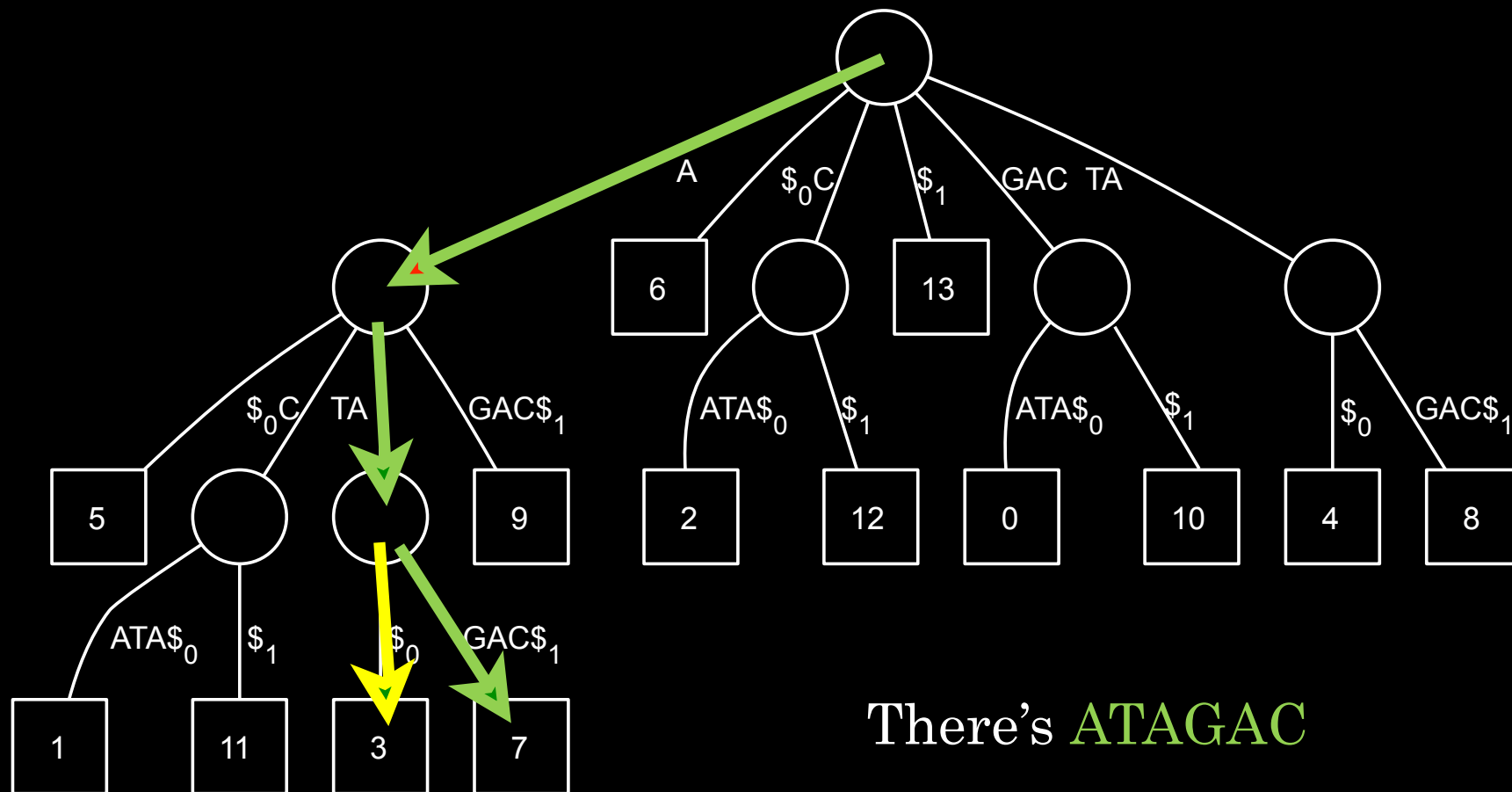
A terminator exists after GAC!

Read #1 ends with GAC

ATAGAC
|||
GACATA

GACATA\$₀ATAGAC\$₁

Let's start by searching for **ATAGAC**



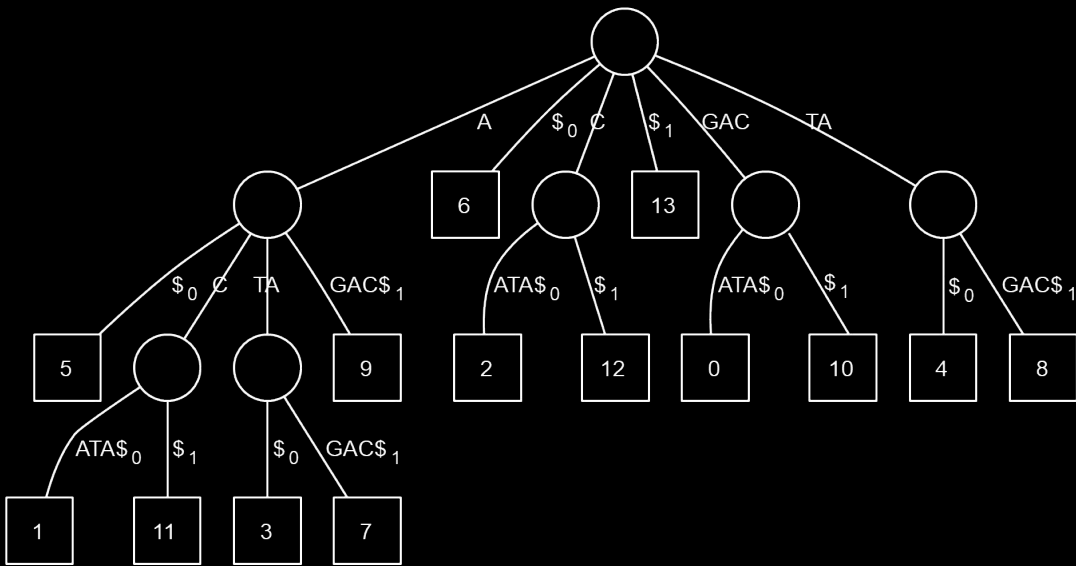
There's ATAGAC

A terminator exists after A+TA!

Read #0 ends with ATA

ATAGAC
 |||
 GACATA

Finding Overlaps - Complexity



Say there are d reads of length n , total length $N = dn$, and $a = \#$ read pairs that overlap

Assume for given read pair we report only the longest suffix/prefix match

Time to build suffix tree: $O(N)$
 ... to walk down red paths: $O(N)$

... to find & report overlaps (green): $O(a)$

Overall: $O(N + a)$

About Those Sequencing Errors....

Exact matching is no longer good enough!

CTCGGCCCTAGG

||| |||||

GGCTCTAGGCC

The solution is to do **dynamic programming**

Finding Overlaps with Dynamic Programming

CTCGGCCCTAGG

||| |||

GGCTCTAGGCC

Use **global alignment** and score function

Match / mismatch scores s
(smaller is better!)

	A	C	G	T	-
A	0	4	2	4	8
C	4	0	4	2	8
G	2	4	0	4	8
T	4	2	4	0	8
-	8	8	8	8	

$$D[i,j] = \min \begin{cases} D[i-1,j] + s(x[i-1], \text{gap}) & \text{Insertion in read } j \\ D[i,j-1] + s(\text{gap}, y[j-1]) & \text{Insertion in read } i \\ D[i-1,j-1] + s(x[i-1], y[j-1]) & \text{Match} \end{cases}$$

How do we force it to find prefix / suffix matches?

Finding Overlaps with Dynamic Programming

How to initialize first row & column
so suffix of X aligns to prefix of Y?

First column gets 0s
(any suffix of X is possible)

First row gets ∞ s
(must be a prefix of Y ; leading
gaps not allowed)

Fill the matrix with scores

Backtrace from last row

The diagram shows a DNA double helix. The left strand is yellow and labeled 'X' at its 5' end. It has the sequence 5'-CTCGGCCTAGG-3'. The right strand is green and labeled 'Y' at its 3' end. It has the sequence 3'-GAGCTACGAC-5'. The strands are connected by horizontal lines representing base pairs: C-G, T-A, C-G, G-C, G-C, C-G, C-G, T-A, A-T, G-C, G-C, and G-C.

Finding Overlaps with Dynamic Programming

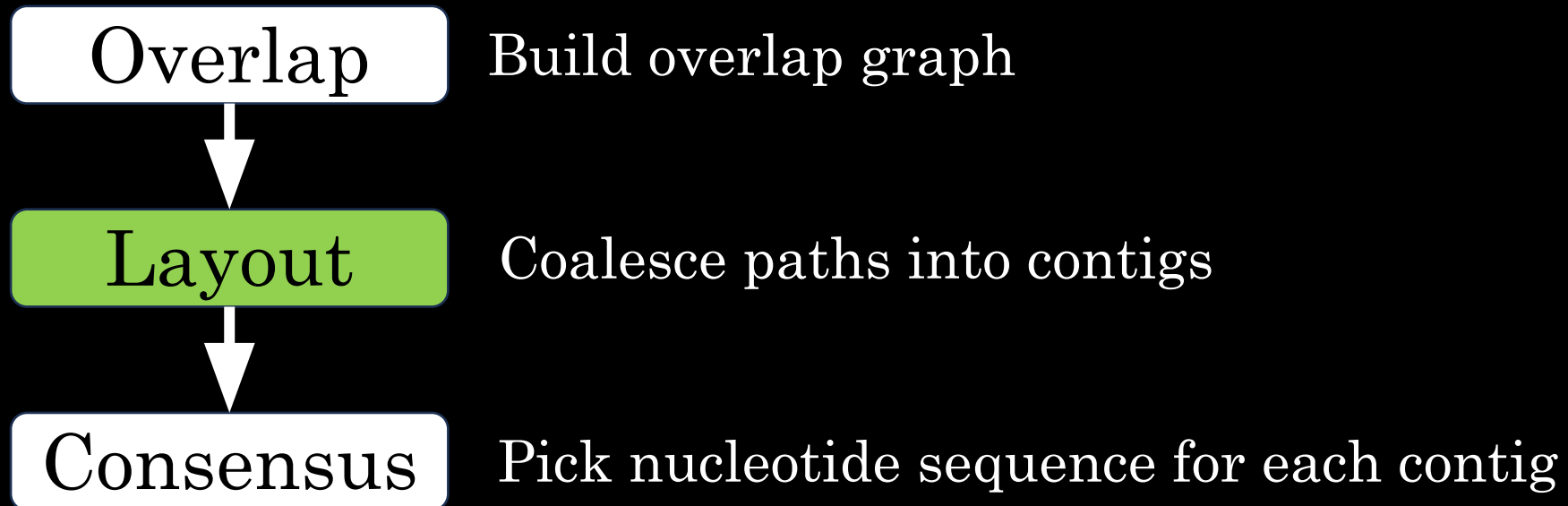
Say there are d reads of length n , total length $N = dn$, and a is total number of pairs with an overlap

# overlaps to try:	$O(d^2)$
Size of each DP matrix:	$O(n^2)$
Overall:	$O(d^2n^2)$, or $O(N^2)$

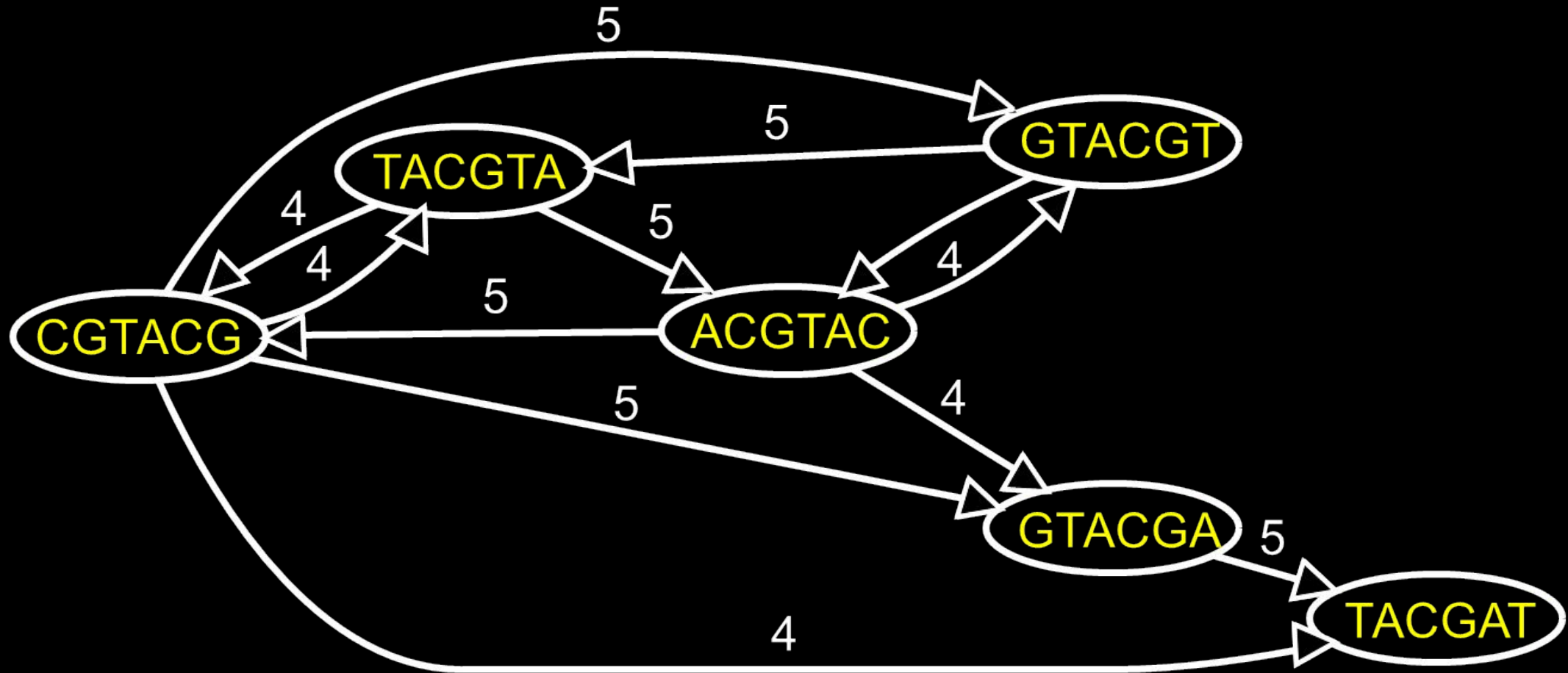
$O(N^2)$ is worse than exact-match suffix tree: $O(N + a)$

Real-world overlappers mix the two; index filters out vast majority of non-overlapping pairs, dynamic programming used for remaining pairs

OLC



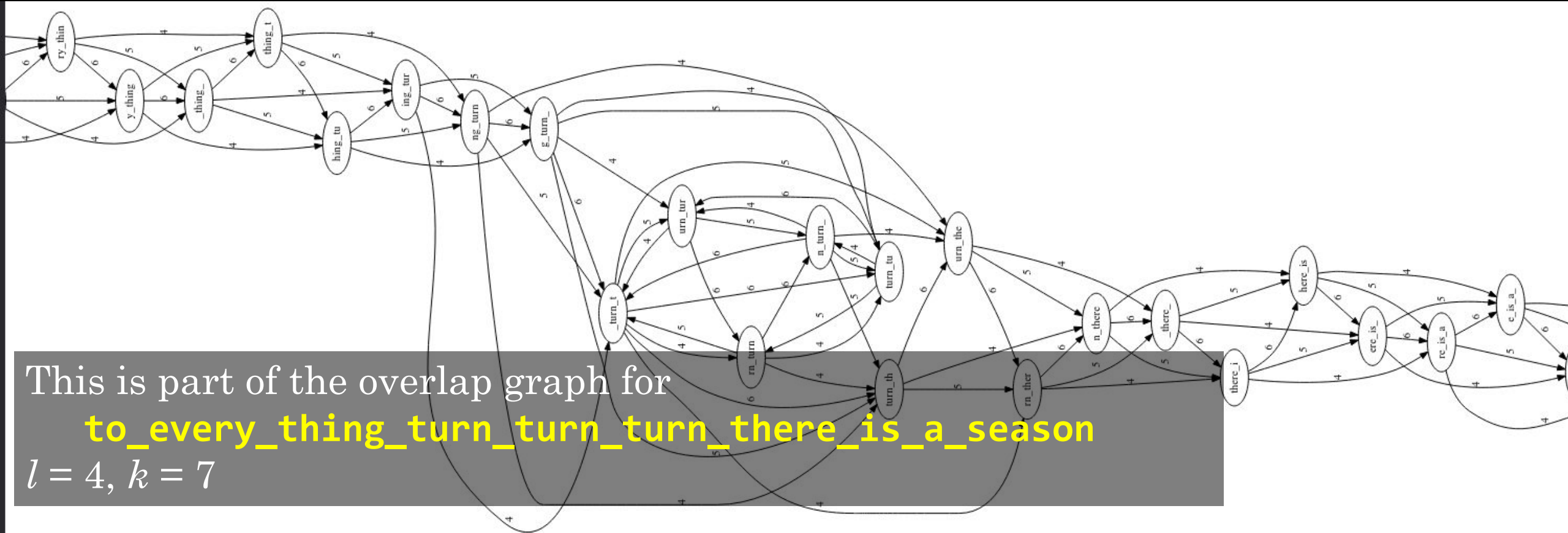
Remember: Overlap Graph



Layout

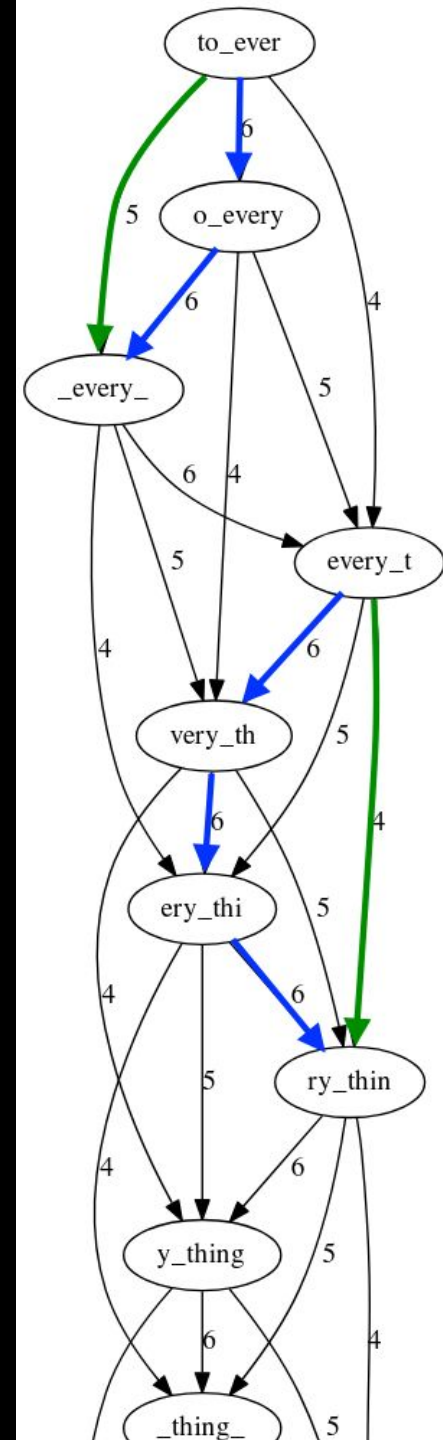
Try to make sense of this.

Overlap graph is big and messy. Contigs don't “pop out” at us.



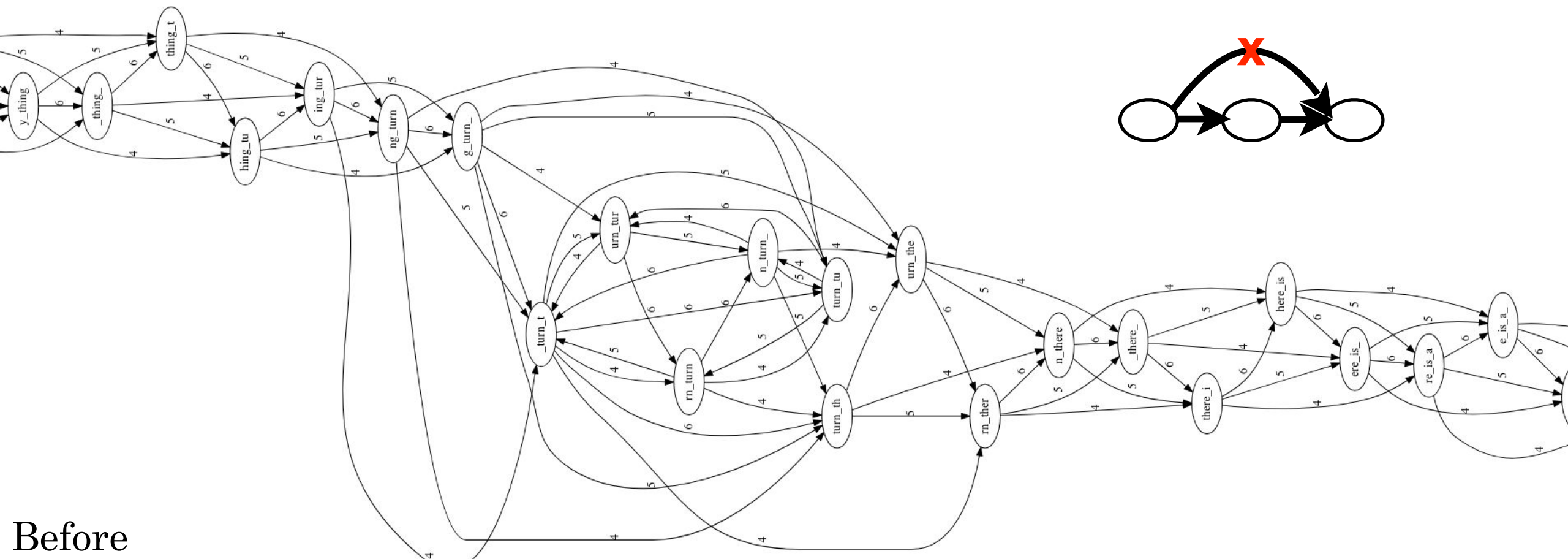
Layout

- We can simplify by removing redundancy from the overlap graph
- Some edges can be inferred (transitively) from other edges
- E.g. **green** edge can be inferred from **blue**



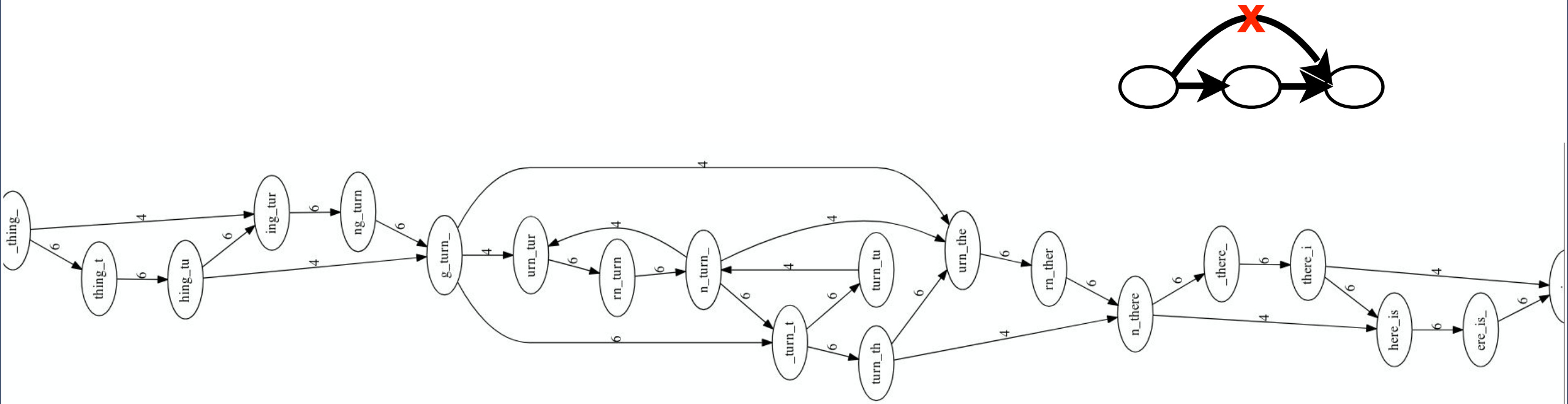
Layout

Remove transitively inferable edges, starting with edges that skip one node:



Layout

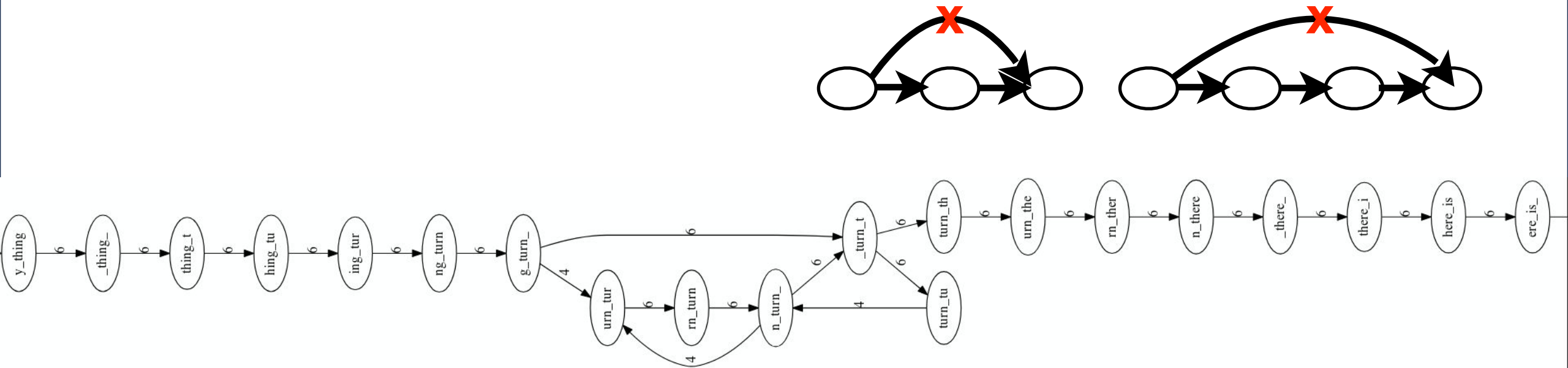
Remove transitively inferable edges, starting with edges that skip one node:



After

Layout

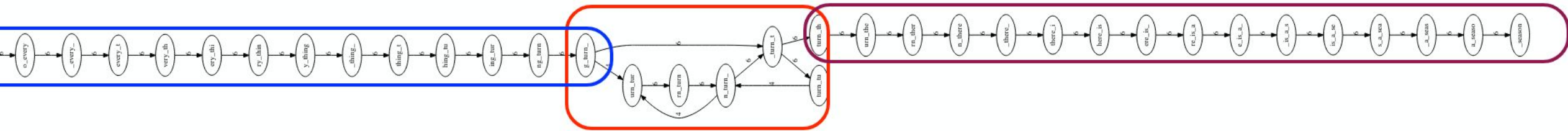
Now remove edges that skip one or two nodes:



After

Layout

Every non-branching stretch is a **contig**



to_every_thing_turn_

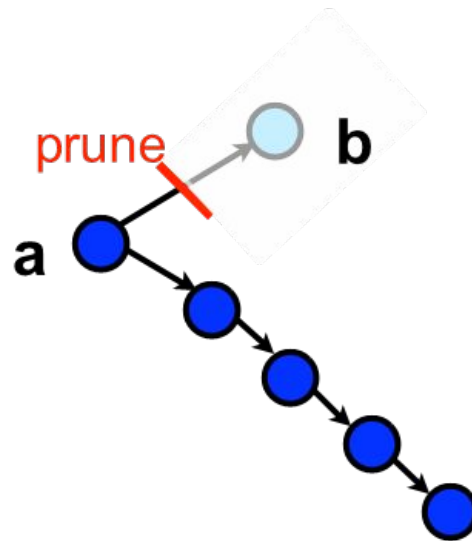
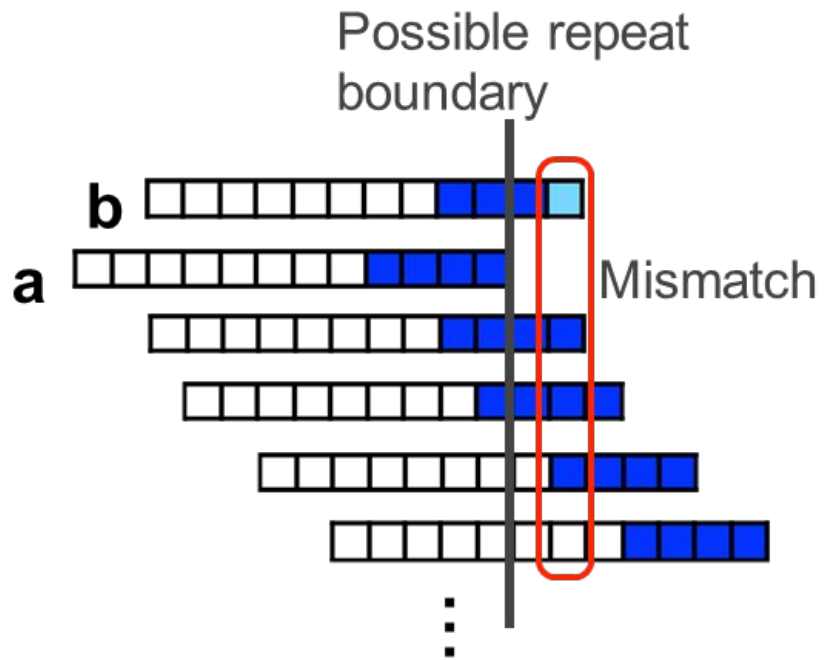
turn_there_is_a_season



Unresolvable branching mess

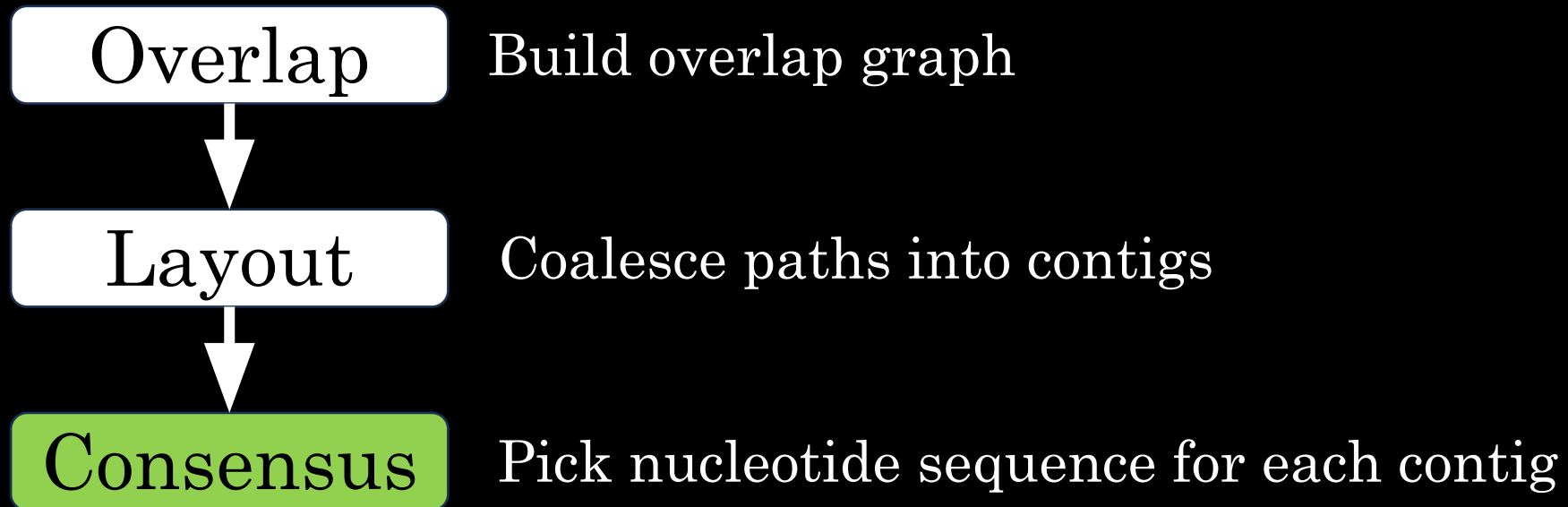
Layout

Spurious weird stuff gets pruned



Forking pattern from (a), early termination at (b). Lies??

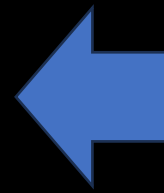
OLC



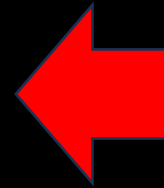
Consensus

TAGATTACACAGATTACTGA TTGATGGCGTAA CTA
TAGATTACACAGATTACTGACTTGGATGGCGTAACTA
TAG TTACACAGATTATTGACTTCATGGCGTAA CTA
TAGATTACACAGATTACTGACTTGGATGGCGTAA CTA
TAGATTACACAGATTACTGACTTGGATGGCGTAA CTA

↓ ↓ ↓ ↓ ↓
TAGATTACACAGATTACTGACTTGGATGGCGTAA CTA

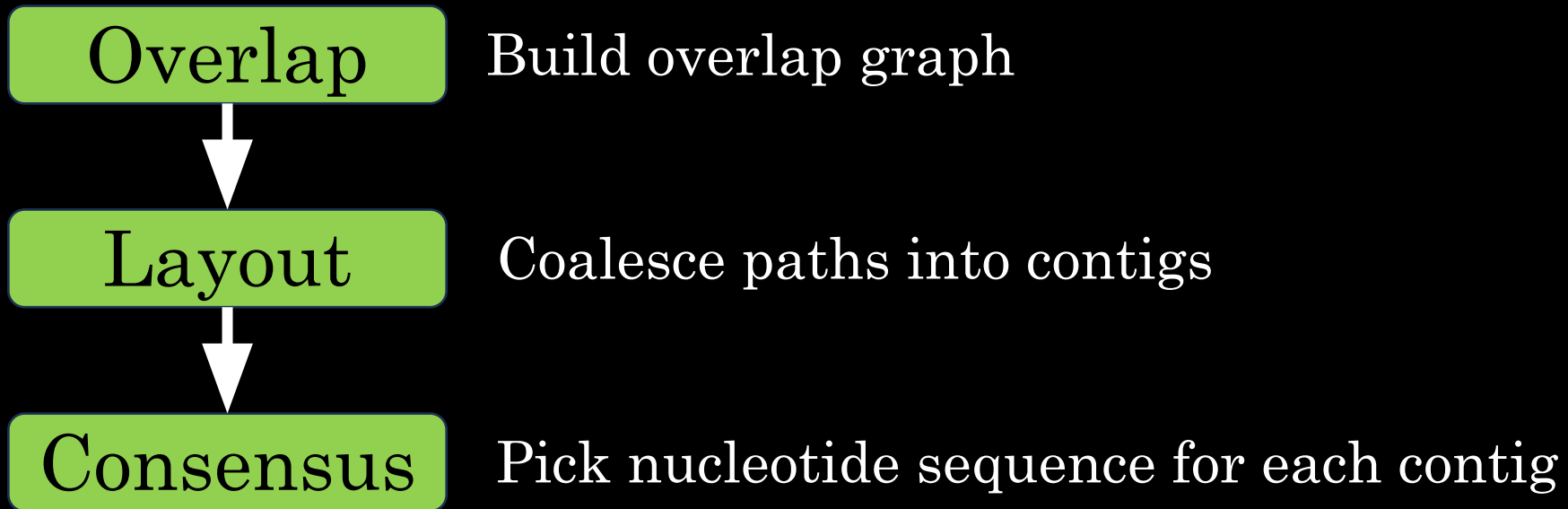


Line up the **reads**
associated with a contig



Take the **consensus**
(majority vote?)

OLC



Problem solved!!!

- **EXCEPT** -

- Building overlap graph is slow. $O(N + a)$, $O(N^2)$
- Overlap graph is **huge**: one node per read, # edges can grow superlinearly with # reads
- Sequencing datasets are ~ 100s of millions or billions of reads

OLC is Generally Not the Approach

- Then why did we just spend a lecture talking about it?
- OLC is more intuitive and covers many of the key concepts we'll see next time
 - Graphs
 - Overlap
 - Weird paths
 - Complex, messy structures
 - Sequencing error